

Monetary Economics

A Self-Contained Course Companion

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How to Read This Book

This is a course companion for a one-semester **monetary economics** course. It is written to be read straight through, like a short textbook, rather than skimmed like a formula sheet.

The book builds monetary theory from first principles. We start in a world without money and ask why a society would adopt it at all, using the overlapping-generations (OLG) framework to show that money can serve as a substitute for a missing record of past transactions — “money is memory.” From that foundation we study commodity money, the inflation tax and the limits of seigniorage, the determination (and indeterminacy) of exchange rates across countries, the institutions and independence of central banks, and finally the mechanics by which a central bank actually changes the money supply through its balance sheet. The early chapters are model-driven and abstract; the later chapters are institutional and concrete. Read them in order: each builds on the language of the one before.

How Each Chapter Is Organized

Each chapter opens with a short *story* in plain English explaining what question we are answering and why we are about to introduce a particular tool. We then formalize the idea, work through the algebra of the model, and read the economics off the diagrams. Definitions and key results are boxed so you can find them quickly when reviewing.

Box Color Code

- **Green: Definition**

— the objects and concepts we build on (fiat money, seigniorage, real money demand, ...).

- **Blue: Theorem / Result**

— the central results of each model.

- Worked **examples**, boxed **remarks**, and emphasis boxes appear throughout.

A Note on Notation

Throughout, a representative agent lives for two periods and consumes c_1 when young and c_2 when old, against an endowment y . We write v_t for the value (purchasing power) of a unit of money at date t , so v_t/v_{t+1} is the gross rate at which money loses or gains value; z denotes the gross money-growth factor and n the gross population-growth factor. a denotes real money balances acquired when young.

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Chapter 1

A Simple Model of Money

Money is strange among the objects economists study. We want bread because eating it gives us utility, and we want a coat because wearing it keeps us warm; the demand for an ordinary good is a demand for the good itself. Money is different. Nobody eats a banknote or wears a coin. We hold money not for its own sake but because it lets us obtain the things we actually want to consume. In the standard threefold description, money serves as a *medium of exchange*, a *unit of account*, and a *store of value*; this chapter takes the first of these as the point of departure and builds the smallest model in which money does something useful.

The model is an overlapping-generations (OLG) economy in which each person is endowed with goods only when young but wishes to consume when old. The economic problem is therefore one of transferring consumption forward in time, from the young to their older selves. We solve it twice. First a benevolent planner allocates the single good directly; this *centralized* solution defines what an efficient outcome looks like — the golden-rule allocation. Then we let people trade on their own, first through bilateral transfers backed by perfect record-keeping and then through intrinsically worthless fiat money. The central result is that money, by serving as a record of past contributions, lets the *decentralized* economy reproduce exactly the planner's allocation. In a slogan that organizes the rest of the chapter: money is memory.

1.1 Overlapping Generations Model

1.1.1 Assumptions

The environment is deliberately spare. Time runs in discrete periods $t = 1, 2, \dots$, without end, and the economy is populated by successive generations whose lives overlap.

- Each individual lives for exactly two periods. She is *young* in the first period of her life and *old* in the second.
- There is a single consumption good, and it is *nonstorable*: a unit produced in period t that is not consumed in period t simply perishes.
- The young are endowed with y units of the good; the old receive no endowment.

- The economy continues for infinitely many periods.

The infinite horizon is not a technical convenience but the heart of the matter. If the economy were known to end at some finite date, the last generation of young would have no reason to give anything to the old — there would be no future selves to reward them for it — and the logic would unravel backward to the present. With no last period, every young generation can be induced to provide for the old in the expectation of receiving the same treatment in turn. We will return to this point repeatedly.

1.1.2 Centralized Solution

Consider first an all-knowing, benevolent planner who allocates the good directly. Let N_t denote the size of the young generation born in period t , and write $c_{1,t}$ for the consumption of a young person and $c_{2,t}$ for that of an old person in period t . The only goods available in period t are the endowments of the current young, since the good cannot be stored and the old produce nothing. The economy's resource constraint in period t is therefore

$$N_t c_{1,t} + N_{t-1} c_{2,t} \leq N_t y,$$

which says that the total consumption of the young and the old alive at t cannot exceed the total endowment of the current young.

We focus throughout on *stationary* allocations, in which a constant population $N_t = N$ for all t consumes the same bundle every period. Dividing the resource constraint by N , the stationary feasible set collapses to

$$c_1 + c_2 \leq y. \tag{1.1}$$

A useful way to read (1.1) is as a lifetime budget for a representative individual: whatever she consumes when young plus whatever she consumes when old must, in a stationary economy, be covered by her single endowment y .

One generation sits outside this logic: the *initial old*, who are old in period 1 and were never young inside the model. A purely selfish initial old generation would want to consume as much as possible, taking $c_{2,1}$ up to y and leaving nothing for intergenerational arrangements to build on. If instead the initial old place some weight on future generations — or if the planner does — the difficulty disappears, and we may treat every generation symmetrically.

The planner chooses the stationary bundle (c_1, c_2) on the frontier $c_1 + c_2 = y$ that the representative individual most prefers. Geometrically, this is the point at which an indifference curve is tangent to the feasible line, illustrated in Figure 1.1.

Definition 1.1: Golden-Rule Allocation

The *golden-rule allocation* is the stationary bundle (c_1^*, c_2^*) that maximizes the representative individual's utility subject to the stationary feasible set $c_1 + c_2 \leq y$. It is the tangency point E between the highest attainable indifference curve and the feasible line, and it is the benchmark against which decentralized outcomes are judged.

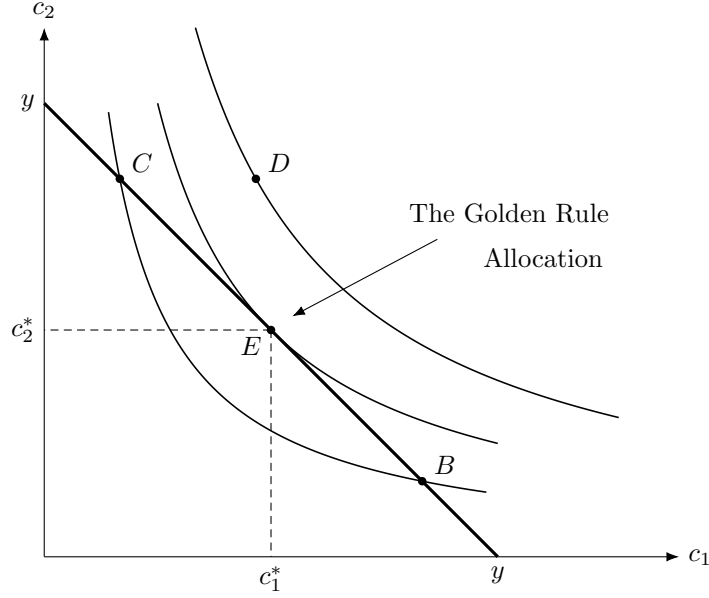


Figure 1.1: The golden-rule allocation in the stationary OLG economy: the planner picks the bundle $E = (c_1^*, c_2^*)$ at which an indifference curve is tangent to the feasible line $c_1 + c_2 = y$; bundles such as D are preferred but infeasible, while feasible bundles such as B and C yield lower utility.

In Figure 1.1, feasible bundles such as B and C lie on the frontier but on lower indifference curves than E , so they deliver less utility; the bundle D lies on a higher indifference curve but outside the feasible set, so it cannot be reached. The golden-rule point E is the best the economy can do.

1.1.3 Decentralized Solution

The planner is a fiction. The real question is whether self-interested individuals, trading on their own, can reproduce the allocation E . Suppose markets are perfectly competitive and, for now, that the economy has a perfect record-keeping system, so that every transfer between generations is observed and remembered.

When young, an individual splits her endowment between current consumption and a transfer φ_t handed to the current old. When old, she consumes whatever the next young generation transfers to her. Her two budget constraints are

$$\begin{aligned} \text{Young: } c_{1,t} + \varphi_t &\leq y, \\ \text{Old: } c_{2,t+1} &\leq \varphi_{t+1}^R, \end{aligned}$$

where φ_t is a choice variable the young set freely, and φ_{t+1}^R is the transfer she receives when old, determined by the *next* young generation and so outside her own control. Define the conversion rate x_{t+1} between a unit given up when young and a unit received when old as

$$x_{t+1} = \frac{\varphi_{t+1}^R}{\varphi_t}.$$

Solving the old-age constraint for $\varphi_t = \varphi_{t+1}^R/x_{t+1} = c_{2,t+1}/x_{t+1}$ at equality and substituting into the youth constraint combines the two into a single lifetime budget,

$$c_{1,t} + \frac{c_{2,t+1}}{x_{t+1}} \leq c_{1,t} + \varphi_t \leq y. \quad (1.2)$$

The conversion rate x_{t+1} is not a free parameter; it is pinned down by market clearing. The transfers received by the old in period t must equal the transfers made by the young in period t . The old of period t each carry a claim φ_{t-1} from when they were young, redeemable at rate x_t , so their total claim — the demand for resources — is

$$\begin{aligned} \text{Demand} &= N_{t-1} c_{2,t} = N_{t-1} (\varphi_{t-1} x_t), \\ \text{Supply} &= N_t (y - c_{1,t}), \end{aligned}$$

where the supply is the goods the current young choose not to consume. Equating demand and supply and solving for x_t ,

$$x_t = \frac{N_t (y - c_{1,t})}{N_{t-1} \varphi_{t-1}} = \frac{N_t (y - c_{1,t})}{N_{t-1} (y - c_{1,t-1})},$$

where the last step uses $\varphi_{t-1} = y - c_{1,t-1}$ from the youth budget at equality. In a stationary equilibrium with constant population and constant consumption, $c_{1,t} = c_1$ for all t and $N_{t-1} = N_t = N$, so the numerator and denominator coincide and

$$x_t = 1.$$

With a unit conversion rate, the lifetime budget (1.2) reduces to

$$c_1 + c_2 \leq y,$$

which is exactly the planner's feasible set (1.1). The decentralized economy with perfect records therefore faces precisely the constraint the planner faced, and the two solutions coincide.

When records are perfect, decentralization is free

With a perfect record-keeping system and a stationary, competitive economy, the conversion rate between youth and old-age consumption is $x_t = 1$, and the representative individual's lifetime budget $c_1 + c_2 \leq y$ is identical to the planner's feasible set. The decentralized allocation reproduces the golden-rule allocation E .

What if records are imperfect? In the centralized economy nothing changes: the planner already knows everything and allocates the good directly. In the decentralized economy, by contrast, the absence of records is fatal. A young person who gives goods to the old today has no way to prove, when she is old, that she ever contributed; the next young generation cannot tell a past contributor from a free rider and has no reason to transfer anything. The only sustainable outcome is *autarky*: no trade between generations, each person consuming only $c_1 = y$ when young and $c_2 = 0$ when old. This is strictly worse than the golden rule. The friction that breaks the decentralized economy is precisely the lack of memory — which

is what fiat money will supply.

1.2 Money is Memory

Instead of transferring goods directly, suppose the government issues an intrinsically useless token — *fiat money* — that the young can sell to the old in exchange for goods. We will see that this token does the job that perfect records did, and for the same reason: holding money is evidence of having contributed in the past.

Definition 1.2: Fiat Money

Fiat money is an object that is

- provided by the government at no cost,
- storable across periods without cost, and
- fully divisible,

and that has no intrinsic value — it yields no utility from consumption and serves no purpose other than as a medium of exchange and store of value.

Suppose the government has issued a fixed stock M of fiat money, and that a representative individual holds m_t units. Let v_t be the real value of one unit of money in period t — the amount of the consumption good it commands — and let p_t be the money price of one unit of the good. The two are reciprocals:

$$v_t = \frac{1}{p_t}.$$

The young now acquire money by selling some of their endowment, and the old finance their consumption by spending the money they accumulated when young. The budget constraints become

$$\begin{aligned} \text{When young:} \quad & c_{1,t} + v_t m_t \leq y, \\ \text{When old:} \quad & c_{2,t+1} \leq v_{t+1} m_t, \\ \implies & c_{1,t} + \frac{v_t}{v_{t+1}} c_{2,t+1} \leq y, \end{aligned}$$

where the third line combines the first two by eliminating m_t . The ratio v_t/v_{t+1} is the price at which old-age consumption trades against youth consumption, and it has a clean interpretation.

Definition 1.3: Real Return of Money

The *gross real return* on money held from period t to period $t + 1$ is v_{t+1}/v_t : one unit of money buys v_t goods when young and v_{t+1} goods when old. Its reciprocal v_t/v_{t+1} is therefore the relative price of old-age consumption that appears in the lifetime budget. The *net* real return is $\frac{v_{t+1}}{v_t} - 1$.

The value of money is determined by market clearing in the money market. The young

supply goods in exchange for money, and the real value of the entire money stock is what the old bring to the market. Demand and supply for goods through the money market are

$$\begin{aligned}\text{Demand} &= N_t (y - c_{1,t}), \\ \text{Supply} &= v_t M_t, \\ \implies v_t &= \frac{N_t (y - c_{1,t})}{M_t},\end{aligned}$$

where the money stock M_t is converted to real terms by multiplying by v_t . Taking the ratio of this expression across two adjacent periods gives the evolution of the value of money,

$$\frac{v_{t+1}}{v_t} = \frac{N_{t+1} (y - c_{1,t+1})}{N_t (y - c_{1,t})} \cdot \frac{M_t}{M_{t+1}}. \quad (1.3)$$

Under stationarity with constant population ($N_{t+1} = N_t$) and a fixed money stock ($M_{t+1} = M_t$), every factor on the right-hand side equals one, so the gross real return on money is

$$\frac{v_{t+1}}{v_t} = 1.$$

The lifetime budget then becomes

$$c_1 + c_2 \leq y,$$

which once again coincides with the planner's feasible set (1.1). Fiat money, holding no intrinsic value whatsoever, has reproduced the golden-rule allocation.

The reason it works is exactly the reason perfect records worked. An old person who holds money is, in effect, holding a verifiable record that she gave up goods when young; the young accept her money because they expect to do the same and be repaid in turn. Where the economy without records collapsed into autarky, money allocates resources across generations by serving as a store of value that doubles as a memory of past contributions. This is the sense in which *money is memory*.

Money is memory

Fiat money has no intrinsic worth, yet in the OLG economy it implements the golden-rule allocation that the recordless decentralized economy could not reach. It does so by acting as a portable, verifiable record of past contributions to the previous generation — it substitutes for the missing record-keeping technology.

It remains to read off the behavior of the price level. Inverting the market-clearing value of money,

$$p_t = \frac{1}{v_t} = \frac{M_t}{N_t (y - c_{1,t})}. \quad (1.4)$$

Equation (1.4) separates what money does from what money is worth. The real allocation — how much each individual consumes when young and old — is determined by preferences and the feasible set and does *not* depend on M_t . A larger money stock raises the price level p_t proportionally but leaves the consumption decision untouched. This is the *neutrality* of money.

Remark (Neutrality is not uselessness).

Neutrality says that the *quantity* of money affects nominal prices but not the real allocation; doubling M_t doubles p_t and halves v_t , leaving every c unchanged. It is easy to misread this as saying money does nothing. It does not say that. The very existence of money is what lets the decentralized economy escape autarky and reach the golden rule in the first place. Money is neutral in its quantity and indispensable in its presence.

1.3 Population Growth

The stationary case set both the population and the money stock constant. We now relax the first of these and let the population grow geometrically,

$$N_t = n N_{t-1}, \quad n > 1,$$

so that each generation is n times larger than the one before. The money stock is still fixed at M .

The resource constraint in period t is unchanged in form: the current young, of whom there are N_t , must feed themselves and the N_{t-1} old. In a stationary allocation $c_{1,t} = c_1$ and $c_{2,t} = c_2$ for all t , so

$$N_t c_1 + N_{t-1} c_2 \leq N_t y \iff c_1 + \frac{1}{n} c_2 \leq y,$$

where the second form divides through by N_t and uses $N_{t-1}/N_t = 1/n$. Population growth relaxes the feasible set: because each old generation is small relative to the young generation supporting it, a given amount of old-age consumption costs the economy less per young person, so the term c_2 enters with the discount factor $1/n$.

Turning to the decentralized economy, the value of money evolves according to (1.3) with $M_{t+1} = M_t$ and $N_{t+1} = nN_t$. Under stationary consumption the factor $(y - c_{1,t+1}) / (y - c_{1,t})$ is one, so the gross real return on money becomes

$$\frac{v_{t+1}}{v_t} = n.$$

Money now *appreciates* in real terms at the population growth rate: a larger young generation arrives each period to buy the fixed money stock from the smaller old generation, bidding up its value. Recall that the lifetime budget is $c_{1,t} + (v_t/v_{t+1}) c_{2,t+1} \leq y$. With $v_{t+1}/v_t = n$ we have $v_t/v_{t+1} = 1/n$, so the stationary lifetime budget becomes

$$c_1 + \frac{1}{n} c_2 \leq y,$$

which is again identical to the feasible set. The conclusion of the stationary case survives population growth intact: the decentralized monetary economy reproduces the planner's allocation, now along a feasible frontier tilted by the growth factor n . Money continues to do the work of memory; growth simply changes the rate at which old-age claims are valued.

Chapter 2

Barter and the Cost of Exchange

The overlapping-generations model of Chapter 1 took the existence of fiat money for granted: a young person hands her goods to an old person and walks away holding intrinsically worthless tokens, confident that some future young person will accept those same tokens in turn. But money is only one of several technologies a society can use to move goods through time and across people. The oldest alternative is *barter* — the direct swap of one good for another, with no intermediating object at all. Why, then, do real economies overwhelmingly choose money?

This chapter answers that question by treating money as a device for *economizing on the cost of trade*. We enrich the environment with many distinct goods and attach an explicit cost to the act of searching for a trading partner and to the act of accepting delivery. Money turns out to win not because it is valuable in itself — by assumption it is not — but because it relaxes the “double coincidence of wants” that cripples barter, and it does so ever more decisively in larger, more specialized economies. We also see the limit of the argument: money’s search-cost advantage can be undone if money itself is expensive to handle.

2.1 A Many-Goods Exchange Economy

We keep the overlapping-generations structure of Chapter 1 but replace the single consumption good with J distinct types of goods.

Environment

There are J different types of goods. When an individual is young she is endowed with y units of *one* type of good; when she becomes old she has no further endowment but wishes to consume *another* type of good. Goods can be stored costlessly, and the economy holds a total stock of fiat money M that is also stored costlessly. Every search for a trading partner — every attempt at a match — carries a cost α .

The crucial friction is that a young person’s good is generically *not* the good her own future self wants to consume, nor the good that the particular old person she meets happens

to want. Trade therefore requires finding the right counterparty, and finding takes costly search. The whole comparison below turns on how many costly searches each exchange technology forces upon a typical individual.

2.2 The Cost of Barter

Under *direct barter*, two parties trade only if each holds exactly what the other wants. This is the classic *double coincidence of wants*: the old person I meet must both have the good I want *and* want the good I have.

Definition 2.1: Double Coincidence of Wants

In direct barter an exchange occurs between two individuals only if each holds the good that the other desires. With J goods, an arbitrary pair of traders satisfies this requirement only for a small fraction of all possible meetings, and the fraction shrinks as J grows.

With J types of goods, the number of ordered “what I have, what I want” combinations across the population is $J^2 - J$ (all J^2 pairs, less the J pairs in which the two goods coincide, which require no trade). Since each match costs α by definition, the average search cost of organizing trade through barter is

$$\text{barter search cost} = \alpha (J^2 - J).$$

The cost grows *quadratically* in the number of goods: doubling the variety of goods in the economy roughly quadruples the search burden of barter.

2.3 The Cost of Monetary Exchange

Now introduce fiat money as a generally accepted means of payment. Trade is broken into two legs. When matched, the young person gives her goods to an old person and in return receives money; later, when old, she gives that money to some young person and in return receives the consumption good she wants.

The key simplification is that money dissolves one half of the double coincidence. A young seller wants *money*, and she does not care which type of old person she meets, because *every* old person carries the one thing she is willing to accept — fiat money. The only coincidence that must still be arranged is that the old person wants the particular good the young person is selling, which happens with probability $\frac{1}{J}$ in a single try.

Why money raises the match probability

Under barter, a successful meeting requires a *two-sided* coincidence of wants. Under monetary exchange only *one* side must coincide, because money is universally accepted. The probability of a successful match is therefore strictly higher with money than with barter.

Each individual now undertakes *two* costly searches over a lifetime — one when young (selling goods for money) and one when old (spending money on goods). Hence the lifetime search cost under monetary exchange averages

$$\text{money search cost} = 2\alpha J,$$

which grows only *linearly* in the number of goods.

2.4 When Does Money Beat Barter?

Comparing the two search costs, money lowers search costs whenever

$$\alpha(J^2 - J) > 2\alpha J \iff J^2 - J > 2J \iff J > 3.$$

Theorem 2.2: Money Dominates Barter in a Complex Economy

Holding the per-search cost α fixed, monetary exchange entails lower total search costs than direct barter if and only if there are more than three types of goods,

$$\alpha(J^2 - J) > 2\alpha J \iff J > 3.$$

Moreover the search-cost advantage of money, $\alpha(J^2 - J) - 2\alpha J = \alpha J(J - 3)$, is increasing in J : money becomes more valuable as the economy grows more complex.

Proof. Divide both sides of $\alpha(J^2 - J) > 2\alpha J$ by $\alpha > 0$ to obtain $J^2 - J > 2J$, i.e. $J^2 - 3J > 0$, i.e. $J(J - 3) > 0$. Since $J \geq 1$, this holds exactly when $J > 3$. The advantage is the gap $\alpha J(J - 3)$, whose derivative in J is $\alpha(2J - 3) > 0$ for all $J \geq 2$, so the gap widens as J rises. $\square$

The economic content is intuitive. In a tiny economy — only two or three goods — finding someone with a matching double coincidence is easy, and the bookkeeping of a two-leg monetary trade is not worth the trouble. But specialization multiplies the number of distinct goods, the double coincidence becomes vanishingly rare, and a universally accepted medium of exchange becomes indispensable. Money is, in this precise sense, a technology whose payoff scales with the sophistication of the division of labor.

2.5 Exchange Costs and the Limits of Money

The search-cost comparison above counts only the cost of *finding* a partner. But every act of trade also carries an *exchange cost* — the cost incurred whenever goods or money are accepted and handled. Here barter enjoys a structural advantage that can, in principle, offset money's gains: barter completes an exchange in a *single* trade (goods for goods), whereas monetary exchange requires *two* trades (goods for money, then money for goods).

Let λ denote the exchange cost of accepting goods, per person, and let λ_M denote the additional exchange cost associated with accepting money. An exchange cost is incurred each time goods or money are accepted.

Definition 2.3: Exchange Cost

The *exchange cost* is the cost borne whenever a party accepts delivery of goods or of money in a trade. We write λ for the cost of accepting goods and λ_M for the cost of accepting money.

Under barter, a person trades only once in a lifetime, and only when old: a young person's consumption comes entirely from her own endowment, so she has no reason to trade while young. She accepts delivery of goods exactly once, so her lifetime exchange cost is

$$\text{barter exchange cost} = \lambda.$$

Under monetary exchange, the same person accepts money once (when young, in exchange for her goods) and accepts goods once (when old, in exchange for her money). Her lifetime exchange cost is therefore

$$\text{money exchange cost} = \lambda + \lambda_M.$$

Remark (The trade-off in one line).

Money buys a lower *search* cost ($2\alpha J$ instead of $\alpha(J^2 - J)$) at the price of a higher *exchange* cost ($\lambda + \lambda_M$ instead of λ). Money is the superior technology only when its search-cost saving exceeds the extra exchange cost λ_M of handling the second trade. A money that is cheap to find a partner for but expensive to accept — a clumsy or insecure money — can lose this comparison even in a large economy.

Combining the two channels, the total lifetime cost of each technology is the sum of its expected search cost and its exchange cost, as collected in Figure 2.1.

	Expected Search Cost	Exchange Cost	Total Cost
Barter	$\alpha(J^2 - J)$	λ	$\alpha(J^2 - J) + \lambda$
Money	$2\alpha J$	$\lambda + \lambda_M$	$2\alpha J + \lambda + \lambda_M$

Figure 2.1: Total lifetime cost of trade under barter and under monetary exchange, decomposed into expected search cost and exchange cost; money lowers the search cost from $\alpha(J^2 - J)$ to $2\alpha J$ but adds the money-handling cost λ_M .

Reading down the total-cost column of Figure 2.1, money is the cheaper technology overall precisely when

$$2\alpha J + \lambda + \lambda_M < \alpha(J^2 - J) + \lambda, \quad \text{i.e.} \quad \alpha J(J - 3) > \lambda_M.$$

This sharpens the earlier result. The bare search-cost comparison, $J > 3$, is the special case $\lambda_M = 0$ in which accepting money is free. Once money carries its own handling cost $\lambda_M > 0$, the threshold rises: the economy must be complex enough — J large enough that

the search saving $\alpha J(J - 3)$ clears the hurdle λ_M — before money is worth adopting. The usefulness of any object as a medium of exchange thus depends jointly on how badly the economy needs to escape the double coincidence of wants and on how cheaply that object can itself be exchanged. This same tension between search frictions and handling costs reappears when we turn to commodity money in Chapter 3.

Chapter 3

Commodity Money

The fiat money of Chapter 1 is, by construction, intrinsically worthless: a piece of paper is valued in equilibrium only because everyone expects everyone else to accept it tomorrow. Historically, however, the first monies were not worthless at all. Gold, silver, salt, cattle, and cigarettes have all circulated as money, and each of them is also a *thing* — a commodity with a use of its own. This is the defining feature of *commodity money*: unlike fiat money, it carries an intrinsic value, so its owner always has an outside option, namely to stop trading it and simply consume it.

This chapter places commodity money inside the same overlapping-generations machinery we built for fiat money. We will find that the equilibrium price of the commodity differs from the fiat case in exactly one respect: its intrinsic value sets a *floor* below which it will not trade. That single inequality reshapes two conclusions of the fiat model. First, the quantity theory of money no longer links the price level cleanly to the total stock of the commodity. Second, although both regimes deliver the *same* consumption possibilities to every future generation, the commodity regime is wasteful: it ties up a genuinely useful resource to do a job that an intrinsically worthless token could do just as well. We make each of these claims precise in turn.

3.1 The Commodity-Money Environment

We keep the demographic and technological structure of the fiat economy and change only the asset. In each period N young agents are born, each endowed with y units of the consumption good when young and nothing when old. Agents live two periods and care about consumption in both, so a young agent who wishes to consume when old must carry purchasing power forward. In the fiat economy that carrier was paper money; here it is a commodity.

Definition 3.1: Commodity Money

Commodity money is a good that serves as a medium of exchange while also possessing an *intrinsic value* — a payoff its holder can obtain by consuming it instead of trading it. Write $\tilde{v}$ for the intrinsic value of one unit of the commodity, measured in units of the consumption good. Fiat money is the special case $\tilde{v} = 0$.

Let the commodity be gold. The total stock is M^g units, held initially by the initial old, who receive $m^g = M^g/N$ units each.¹ Let v_t^g denote the *trading value* of one unit of gold at date t , again measured in units of the consumption good, so that $1/v_t^g$ is the gold price of one consumption good. The intrinsic value $\tilde{v}$ and the trading value v_t^g are distinct objects: the former is a fixed parameter of the technology, the latter an endogenous price determined in equilibrium.

3.2 The Budget Constraint and Market Clearing

A representative agent born at date t chooses how much gold m_t^g to carry from youth into old age. When young she splits her endowment between consumption and gold purchases; when old she sells the gold and consumes the proceeds. Her two-period problem is governed by

$$\begin{aligned} \text{when young: } & c_{1,t} + v_t^g m_t^g \leq y, \\ \text{when old: } & c_{2,t+1} \leq v_{t+1}^g m_t^g. \end{aligned}$$

The first line says spending plus gold purchases cannot exceed the endowment; the second says old-age consumption is financed entirely by selling the gold carried over, valued at next period's trading price. Solving the second constraint for $m_t^g = c_{2,t+1}/v_{t+1}^g$ and substituting into the first gives the consolidated lifetime budget constraint

$$c_{1,t} + \frac{v_t^g}{v_{t+1}^g} c_{2,t+1} \leq y. \quad (3.1)$$

The price of future consumption in terms of present consumption is the gross rate of return on holding gold reversed, v_t^g/v_{t+1}^g . When the trading value of gold is rising ($v_{t+1}^g > v_t^g$), carrying gold forward is rewarded and future consumption is cheap. This is structurally identical to the fiat budget constraint of Chapter 1, with the value of gold playing the role formerly played by the value of fiat money.

Remark (Same form, different asset).

Constraint (3.1) is the same lifetime budget line we derived in the fiat economy. The price ratio v_t^g/v_{t+1}^g inherits all the comparative statics of the fiat ratio v_t/v_{t+1} . What is genuinely new in the commodity case is not the agent's optimization but a side constraint on the equilibrium price, developed in Section 3.3.

We now impose market clearing. The young as a group wish to acquire gold using their unspent endowment, $N(y - c_{1,t})$ units of the consumption good. The old as a group supply the entire stock, $Nm^g = M^g$ units of gold, whose total trading value is $v_t^g M^g$. Equating the value of gold demanded to the value of gold supplied,

$$N(y - c_{1,t}) = v_t^g (N m^g) = v_t^g M^g.$$

Restricting attention to a *stationary* equilibrium, in which quantities and prices are constant over time, $c_{1,t} = c_1$, $v_t^g = v^g$, and $m_t^g = m^g$ for all t , the clearing condition pins down the

¹As in the fiat model, the initial old are the only generation that begins life holding the asset; every later generation must acquire it by selling goods when young.

trading value of gold:

$$v^g = \frac{N(y - c_1)}{M^g}. \quad (3.2)$$

This is the commodity-money analogue of the fiat pricing equation: the value of one unit of the medium of exchange is the total real saving of a young generation divided by the stock of the medium. Doubling the stock M^g halves the trading value, holding real saving fixed — the seed of the quantity theory, to which we return.

3.3 The Intrinsic-Value Floor

Here the commodity case parts ways with the fiat case. Because gold has an intrinsic value $\tilde{v}$, no one will ever trade away a unit of gold for less than $\tilde{v}$ worth of goods: doing so is dominated by simply consuming the gold. Gold therefore circulates as money if and only if its trading value is at least its intrinsic value.

When commodity money circulates

Gold is used as a medium of exchange if and only if its trading value weakly exceeds its intrinsic value,

$$v^g = \frac{N(y - c_1)}{M^g} \geq \tilde{v}.$$

If this inequality fails, gold exits circulation and is consumed rather than traded.

What happens when the candidate trading value (3.2) falls short of $\tilde{v}$? Suppose $v^g < \tilde{v}$. Then the initial old prefer to consume gold rather than trade it away at a loss. They do not, however, consume *all* of it. As gold is consumed, the stock M^g shrinks, and by (3.2) a smaller stock raises the trading value v^g . As long as $v^g < \tilde{v}$ the consumption of gold continues and the trading value keeps climbing. The process halts precisely when the trading value has risen to meet the intrinsic value, $v^g = \tilde{v}$. From that point on, no further gold is consumed and the remainder circulates as money. In equilibrium, therefore, the trading value of a commodity money can never settle below its intrinsic value:

$$v^g \geq \tilde{v} \quad \text{whenever gold is used as a medium of exchange.}$$

The intrinsic value acts as an absorbing floor: the market self-adjusts the *quantity* of gold in monetary use until the value of the gold that remains is at least $\tilde{v}$.

3.4 The Quantity Theory Revisited

Because commodity money can be consumed at the margin, the quantity theory of money need not hold in the form it took for fiat money. Recall the fiat statement.

Quantity theory of money (fiat case)

Take two economies identical in every respect except that one holds twice the stock of fiat money. Then the price level is twice as high — equivalently, the value of money is half as high — in the economy with the larger stock. Prices adjust to the

stock of money.

Now run the same comparison with gold, in two cases.

Case 1: gold is never consumed. If gold serves purely as a commodity money and is never consumed, the comparison is identical to the fiat case. The economy with twice the gold stock has prices twice as high, exactly as (3.2) predicts when M^g doubles and $N(y - c_1)$ is held fixed. The quantity theory holds.

Case 2: gold is consumed at the margin. Suppose instead that in both economies gold is consumed up to the point where its trading value just equals its intrinsic value, $v^g = \tilde{v}$. By the floor mechanism of Section 3.3, the economy that began with twice the gold stock simply consumes more gold until its trading value, too, equals $\tilde{v}$. After this adjustment the *quantity of gold actually used as money* is the same in both economies, and so is the price level. The intrinsic value $\tilde{v}$ has imposed a *minimum* on the trading value of gold — equivalently, a *maximum* on the nominal price level — that prevents the larger-stock economy from having higher prices.

The upshot is a careful restatement of the quantity theory for commodity money.

Quantity theory of money (commodity case)

Compare two economies that differ only in their gold stock, with gold consumed at the margin in both.

- In terms of the *initial* stocks of gold, the quantity theory *fails*: the economy with the larger initial stock does not have a proportionately higher price level, because the floor $v^g \geq \tilde{v}$ caps prices.
- In terms of the stocks of gold *actually used as money*, the quantity theory *holds* — but for a reversed reason. Here the stock of monetary gold adjusts to the price level, rather than the price level adjusting to the stock of money.

The reversal of causation in the second bullet is the heart of the matter. Under fiat money the price level chases an exogenously fixed stock; under commodity money with marginal consumption the monetary stock chases a price level anchored at $\tilde{v}$.

3.5 Distribution of Wealth

Because the trading value of a commodity money equals or exceeds its intrinsic value whenever it is used as a medium of exchange, *what* serves as money in an economy carries consequences for the distribution of wealth, not merely for the price level. The owners of the monetary commodity enjoy a trading value $v^g \geq \tilde{v}$ on their holdings; that premium over intrinsic value is theirs only so long as the commodity retains its monetary role.

Example (Replacing gold with fiat money).

Suppose an economy runs on commodity money with $v^g > \tilde{v}$, and it switches to a fiat money system. With gold demonetized, no one trades for it above its intrinsic value, so its price collapses to $v^g = \tilde{v}$. The wealth of gold holders falls by the lost monetary

premium $v^g - \tilde{v}$ per unit. This is why owners of gold — or of any candidate commodity money — take a keen interest in which medium of exchange their economy adopts: a change of monetary regime is a transfer of wealth toward or away from them.

3.6 The Inefficiency of Commodity Money

We close by comparing the two regimes on welfare grounds, starting from a common microeconomic foundation. The striking result is that commodity money is *inefficient*: it can be replaced by fiat money in a way that makes someone strictly better off and no one worse off.

First, the two regimes offer identical consumption possibilities to every generation. The lifetime budget constraint (3.1) has the same form whether the carrier of value is gold or paper; the set of feasible consumption pairs (c_1, c_2) available to any young agent is the same in both economies. With regard to all future generations, then, the commodity regime confers no advantage and no disadvantage relative to fiat money.

Commodity money is inefficient

Switching from commodity money to fiat money makes the *initial old* strictly better off while leaving every future generation exactly as well off as before. A fiat system supports the same trading patterns as a commodity system while freeing up the intrinsically valuable commodity for its nonmonetary uses.

The gain accrues to the initial old. In a fiat regime they sell their fiat money for precisely the same amount of the consumption good they would have obtained by selling gold — the medium of exchange does the same monetary work either way. But on top of that, with gold demonetized they may now *consume* their entire stock of gold, harvesting its intrinsic value $\tilde{v}$ per unit. Their consumption and utility are therefore strictly higher under fiat money, and this improvement is achieved without lowering the welfare of any later generation.

The intuition is one of resource cost. Under commodity money, a resource that has genuine intrinsic value — gold, with its uses in jewelry, electronics, dentistry, and the like — is tied up to perform the function of a medium of exchange. Under fiat money the same monetary services are provided by an intrinsically worthless token, leaving the valuable commodity free to be put to its productive nonmonetary uses. The lesson generalizes well beyond gold: any economy that monetizes a useful good pays a real resource cost for its money, a cost a fiat system avoids. We carry this efficiency comparison forward when we examine inflation and the optimal provision of money in Chapter 4.

Chapter 4

Inflation and Seigniorage

In the overlapping-generations model of Chapter 1, a constant stock of fiat money supported a clean intertemporal trade: the young sold goods for money, the old spent the money back, and a unit of money held its value from one period to the next. We now let the government *print*. Once the money stock grows, the value of money no longer stands still, and the same currency that smoothed exchange becomes a vehicle for taxation. This chapter studies that double role.

The central economic idea is simple to state and surprisingly far-reaching. An increasing fiat money stock acts as an implicit proportional tax on money holdings. Faced with money that loses value over time, individuals economize on their holdings; in doing so they forgo some of the very benefits that fiat money provides, and they end up at a lower level of utility than a stable currency would allow. We trace this distortion through the budget set, ask how much real revenue the government can actually extract from printing money (the answer is: a bounded amount, because the tax base shrinks when the tax rate rises), and finally contrast the inflation tax with a lump-sum tax that raises the same revenue without distortion. Throughout we reuse the equilibrium machinery of Chapter 1; the only new ingredient is a money-growth factor z .

4.1 Basic Model

We keep every feature of the overlapping-generations environment and add a growing money supply. Let the stock of fiat money expand at the constant gross rate z ,

$$M_t = z M_{t-1}, \quad z > 1,$$

so that z is the gross growth factor of the money stock. It will be convenient to express the new money issued in period t as a fraction of the new total: from $M_t = z M_{t-1}$ we have $M_{t-1} = M_t/z$, hence

$$M_t - M_{t-1} = \left(1 - \frac{1}{z}\right) M_t.$$

Suppose the newly created money is distributed evenly to the current old as a lump-sum

transfer, worth a_t units of the consumption good per old person:

$$a_t = \frac{v_t(M_t - M_{t-1})}{N_{t-1}} = \frac{v_t \left(1 - \frac{1}{z}\right) M_t}{N_{t-1}}.$$

In words, the freshly printed money enters the economy as a lump-sum subsidy handed to the old.

Definition 4.1: Real Money Demand

The *real demand for money* of a young person is the real value of the money balances she chooses to carry into old age,

$$v_t m_t = y - c_{1,t},$$

i.e. the part of her endowment she does not consume when young. Aggregated across the young it equals the real value of the money stock, $v_t M_t = N_t(y - c_{1,t})$.

The budget constraint with transfers

With a lump-sum transfer to the old, the two-period problem of a representative young person born at t is

$$\text{When young: } c_{1,t} + v_t m_t \leq y,$$

$$\text{When old: } c_{2,t+1} \leq v_{t+1} m_t + a_{t+1},$$

$$\implies c_{1,t} + \frac{v_t}{v_{t+1}} c_{2,t+1} \leq y + \frac{v_t}{v_{t+1}} a_{t+1}.$$

The transfer relaxes the lifetime constraint by $\frac{v_t}{v_{t+1}} a_{t+1}$, the present value of the old-age subsidy.

The money market clears exactly as before: aggregate real money demand equals the real value of the money stock,

$$N_t (y - c_{1,t}) = v_t M_t \implies v_t = \frac{N_t (y - c_{1,t})}{M_t}.$$

Taking the ratio of this expression at $t+1$ and t under a stationary allocation (so $c_{1,t} = c_1$ for all t) gives the gross rate of return on money,

$$\frac{v_{t+1}}{v_t} = \frac{N_{t+1}/N_t}{M_{t+1}/M_t} = \frac{n}{z},$$

where $n = N_{t+1}/N_t$ is the population growth factor and $z = M_{t+1}/M_t$ the money growth factor. The return on money is the ratio of the growth of money *demand* to the growth of money *supply*.

Inflation as a fall in the return on money

Specialize now to a constant population, $n = 1$, with a growing money supply, $z > 1$. Then

$$\frac{v_{t+1}}{v_t} = \frac{1}{z} \quad \Longleftrightarrow \quad \frac{v_{t+1}}{v_t} - 1 = \frac{1}{z} - 1 < 0.$$

The value of money falls over time. Equivalently, since the price level is $p_t = 1/v_t$, prices rise: more dollars chase the same quantity of goods, so each dollar buys less. This is inflation, and its real cost is that money offers a rate of return below the one a stable currency ($z = 1$) would deliver.

Inflation lowers the return on money

When the money supply grows faster than money demand ($z > n$), the gross return on money $v_{t+1}/v_t = n/z$ is less than one: money loses value over time. With constant population this return is exactly $1/z$. The faster the printing, the worse the return.

Using $v_t/v_{t+1} = z$ and stationarity, the lifetime budget constraint collapses to

$$c_1 + z c_2 \leq y + z a.$$

Inflation has reshaped the budget set in two ways, both visible in Figure 4.1. First, the budget line is *flatter*: the slope (in absolute value) is now $1/z < 1$, so to obtain one unit of old-age consumption a person must give up z units of youthful consumption rather than one. This is precisely the lower rate of return offered by money when new money is being created. Second, the horizontal intercept is $y + za$ rather than y , because lifetime resources now include both the endowment and the subsidy.

Where the subsidy really comes from

It is tempting to read the outward shift of the budget line as a free lunch, but it is not. Although the government can create fiat money out of thin air, the real value of the subsidy must come from *somewhere*. Printing intrinsically worthless pieces of paper does not magically enlarge what the economy can produce. The total number of goods is fixed at the aggregate endowment $N_t y$, so any gift to the old must be paid for by losses borne by the old themselves or by others.

The genuine resource constraint — the *feasible set* for stationary allocations — is unchanged by the printing press. Adding up the consumption of the N_t young and the N_{t-1} old in period t ,

$$N_t c_{1,t} + N_{t-1} c_{2,t} \leq N_t y \quad \Longleftrightarrow \quad c_1 + c_2 \leq y$$

under stationarity and constant population. The feasible-set line starts at y on the vertical axis and has slope -1 , steeper than the budget line.

Theorem 4.2: Equilibrium lies on the feasible-set boundary

The stationary monetary-equilibrium bundle (c_1^*, c_2^*) lies *on* the feasible-set line $c_1 + c_2 = y$, where the feasible-set line crosses the (flatter) budget line.

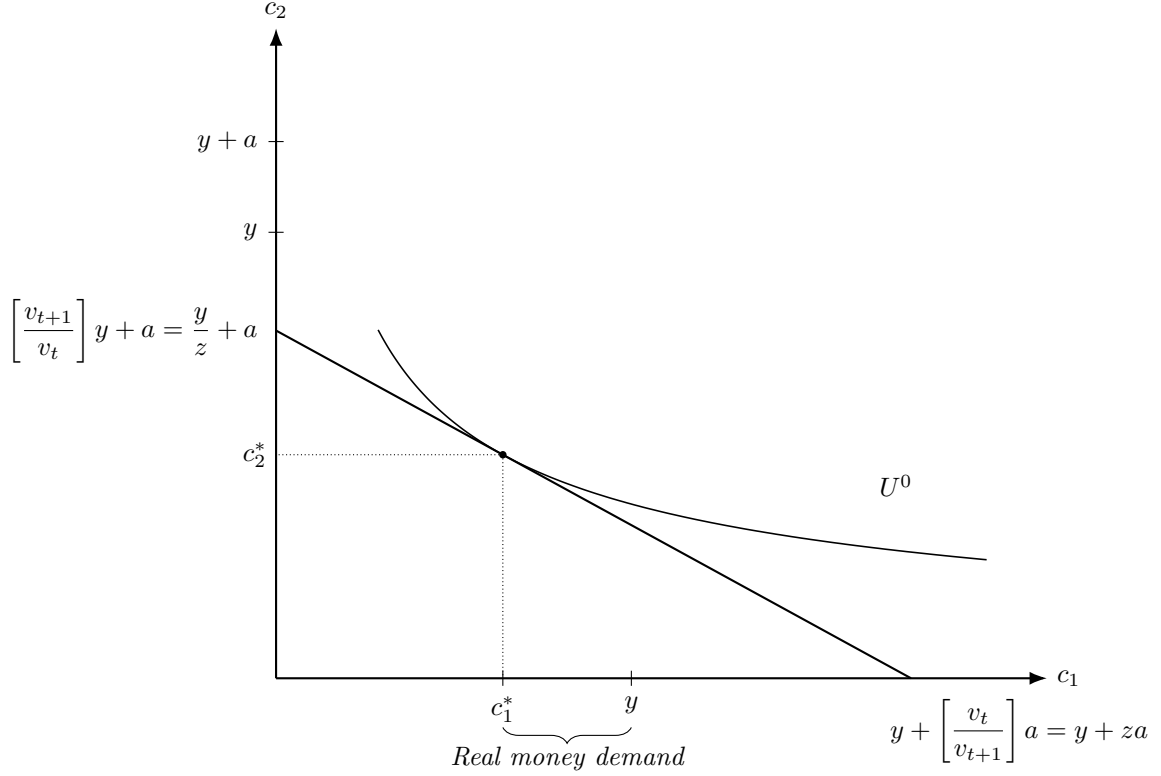


Figure 4.1: Stationary monetary equilibrium under inflation: the flatter budget line $c_1 + zc_2 \leq y + za$ is tangent to an indifference curve at (c_1^*, c_2^*) , and the brace marks the resulting real money demand $y - c_1^*$.

Proof. If (c_1^*, c_2^*) lay strictly inside the feasible set, some goods would be thrown away, which no utility-maximizer would do; if it lay outside, people would consume more goods than exist, which is impossible. Hence the equilibrium bundle must sit on the boundary. The algebraic verification is given in Section 4.3. $\square$

This crossing is shown in Figure 4.2. Point A, where the feasible-set line is tangent to a higher indifference curve, is preferred by every future generation over the inflationary equilibrium (c_1^*, c_2^*) , because it lies on a higher indifference curve. Moreover, second-period consumption is higher at A than at (c_1^*, c_2^*) , so even the initial old — who care only about c_2 — prefer A. Inflation makes *everyone* worse off.

The mechanism is a tax. The inflation tax on money balances induces future generations to *reduce* their money demand $y - c_1$ below the level they would hold in the efficient equilibrium. This drop in demand for fiat money lowers the value of the initial balances held by the initial old, reducing their utility as well. Because the inflation tax is a *proportional* tax — the same rate falls on everyone holding money — it distorts the incentive to hold real balances, and through that the incentive to supply labor. Inflation can therefore affect *aggregate output*, not merely the *timing* of consumption.

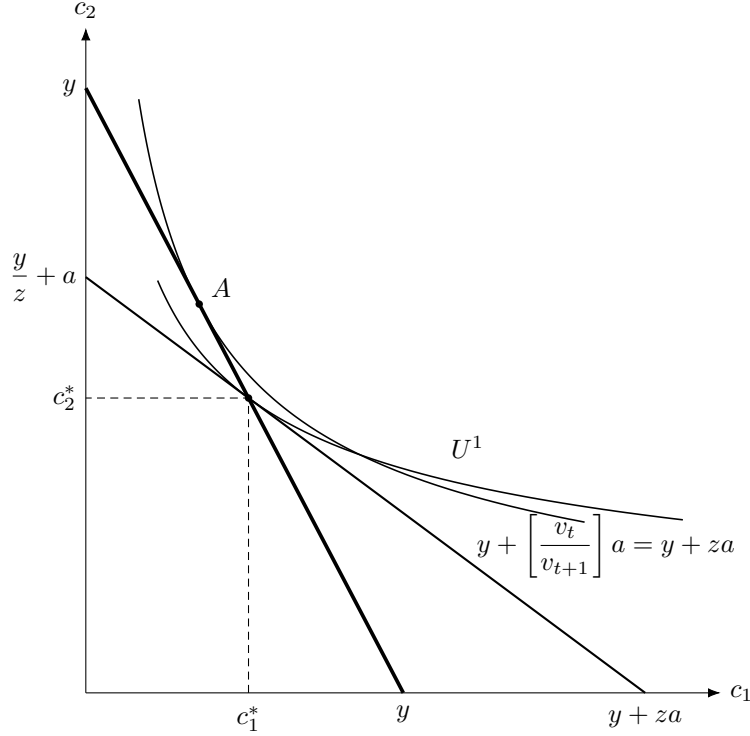


Figure 4.2: The inflationary equilibrium (c_1^*, c_2^*) sits where the steeper feasible-set line meets the flatter budget line; the no-inflation allocation A lies on a strictly higher indifference curve and Pareto-dominates it.

Loss and gain balance for the marginal holder

For an individual who carries balances from $t - 1$ to t , the real loss from inflation — the erosion in the value of the money she holds — is

$$\frac{M_{t-1}(v_{t-1} - v_t)}{N} = \frac{M_{t-1} v_t (z - 1)}{N},$$

using $v_{t-1} = z v_t$ (constant population, $v_t/v_{t+1} = z$). The gain she receives is the lump-sum transfer,

$$a_t = \frac{(M_t - M_{t-1}) v_t}{N} = \frac{M_{t-1} v_t (z - 1)}{N}.$$

The two are equal: the transfer just compensates the holder for the inflation she suffers. Inflation is therefore a pure redistribution at the margin — it transfers no net resources — yet it still destroys welfare, because the *anticipation* of the tax makes people hold too little money in the first place. The deadweight loss is the distortion, not a net transfer.

4.2 The Limits to Seigniorage

The government can print any number of dollars, but the value of those dollars shrinks the more it prints. Real government revenue is therefore bounded by the real value of the money stock; seigniorage is not an unlimited source of funds.

Definition 4.3: Seigniorage

Seigniorage is the real revenue a government earns by issuing new money. In period t it equals the real value of the newly printed money,

$$(M_t - M_{t-1}) v_t = \left(1 - \frac{1}{z}\right) v_t M_t.$$

Here $v_t M_t$ is the real value of the money stock — the *tax base* — and $(1 - \frac{1}{z})$ is the fraction of that base captured by the government — the inflation *tax rate*.

The catch is that the base depends on the rate. When z rises, each person reduces the real balances she holds, precisely to limit the goods she loses to the government through inflation. Lower real money demand means a smaller $v_t M_t$, so the tax base shrinks as the money-growth rate increases. Revenue is the product of a rising tax rate $(1 - \frac{1}{z})$ and a falling tax base $v_t M_t$, so it cannot grow without bound — and beyond some point it falls. Figure 4.3 shows the mechanism behind the shrinking base: raising the money-growth factor from the constant-money benchmark $z = 1$ to $z > 1$ contracts the real balances each young household chooses to hold.

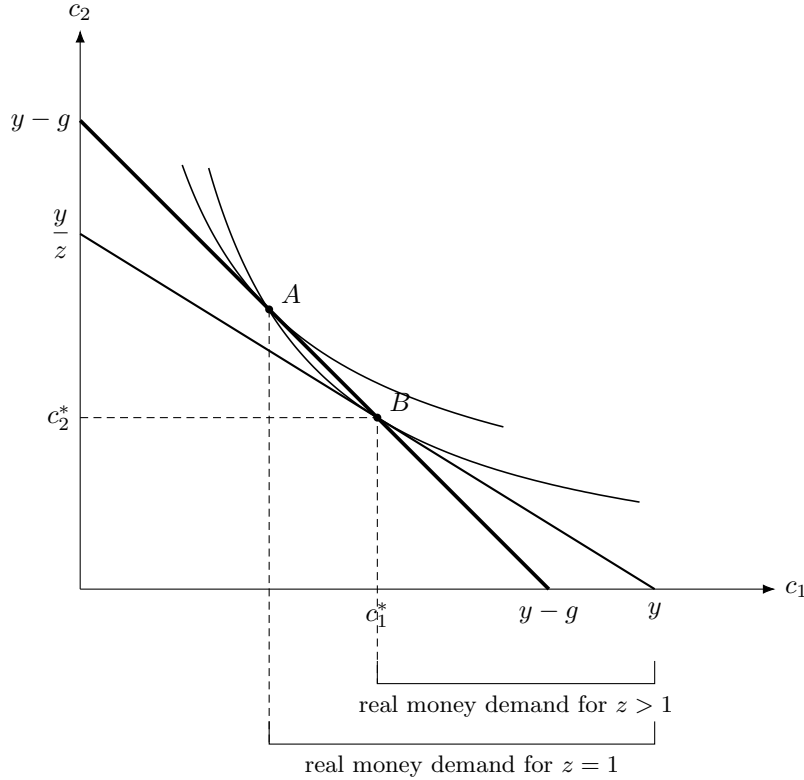


Figure 4.3: Real money demand and the money-growth rate. The grey line (slope -1 , intercepts $y - g$) and the flatter black line (intercepts y/z and y) give the household's optimum at A and B; the two brackets beneath the axis show that real money demand is smaller when $z > 1$ than in the constant-money case $z = 1$ — the mechanism by which a higher inflation rate erodes the seigniorage tax base.

Remark (The inflation Laffer curve).

Just as an income tax rate of 100% would raise no revenue because no one would work, an inflation rate so high that money becomes worthless raises no real revenue because no one holds money. Real-world hyperinflations sit on the wrong side of this curve: the government prints ever faster yet extracts ever less, because the public flees the currency.

4.3 Equilibrium Consumption and the Feasible Set

We now verify algebraically the claim made in Figure 4.2: that all goods are consumed in equilibrium, i.e. the monetary-equilibrium bundle (c_1^*, c_2^*) lies on the feasible-set line. Three equations from the analysis above describe the stationary monetary equilibrium — the lifetime budget constraint, the definition of the subsidy, and money-market clearing:

$$\begin{aligned} c_1^* + \frac{z}{n} c_2^* &= y + \frac{z}{n} a, \\ a &= \frac{\left(1 - \frac{1}{z}\right) v_t M_t}{N_{t-1}}, \\ v_t M_t &= N_t (y - c_1^*). \end{aligned}$$

Substitute the clearing condition into the subsidy, using $N_t = n N_{t-1}$ so that $N_t/N_{t-1} = n$ and $v_t M_t/N_t = v_t m_t = y - c_1^*$:

$$a = \frac{\left(1 - \frac{1}{z}\right) v_t M_t}{N_{t-1}} = \left(1 - \frac{1}{z}\right) v_t M_t \cdot \frac{n}{N_t} = \left(1 - \frac{1}{z}\right) n (y - c_1^*).$$

Insert this into the lifetime budget constraint:

$$c_1^* + \frac{z}{n} c_2^* = y + \frac{z}{n} \cdot \left(1 - \frac{1}{z}\right) n (y - c_1^*) = y + (z - 1)(y - c_1^*).$$

Collecting terms,

$$c_1^* + \frac{z}{n} c_2^* = y + (z - 1)(y - c_1^*) \implies c_1^* + \frac{1}{n} c_2^* = y.$$

With constant population $n = 1$ this is exactly $c_1^* + c_2^* = y$, the feasible-set line. The growth factors z and the subsidy a have cancelled, confirming that they cannot change what is feasible — only how the equilibrium sits relative to the feasible boundary.

4.4 Growing Population

Now allow the population to grow, $N_t = n N_{t-1}$ with $n > 1$. From money-market clearing $v_t = N_t(y - c_1)/M_t$, the return on money becomes

$$\frac{v_{t+1}}{v_t} = \frac{n}{z},$$

where stationarity ($a_{t+1} = a$, $c_{1,t} = c_1$ for all t) is again assumed. Money demand now grows at the population rate n . To keep the value of money — and hence the price level —

constant, the government must expand the money stock at exactly the rate at which money demand grows, that is, it must set

$$z = n.$$

Printing money at the rate of population growth is non-inflationary, because the extra currency is exactly absorbed by the extra young who want to hold it.

The lifetime budget line has the same algebraic form as before. Substituting $v_t/v_{t+1} = z/n$ and stationarity,

$$c_1 + \frac{z}{n} c_2 \leq y + \frac{z}{n} a.$$

The feasible set, by contrast, is shaped only by the real resources of a growing economy — neither printing money nor the subsidy can change it:

$$c_1 + \frac{1}{n} c_2 \leq y.$$

Note that z and a do not appear: they are nominal and redistributive, not technological.

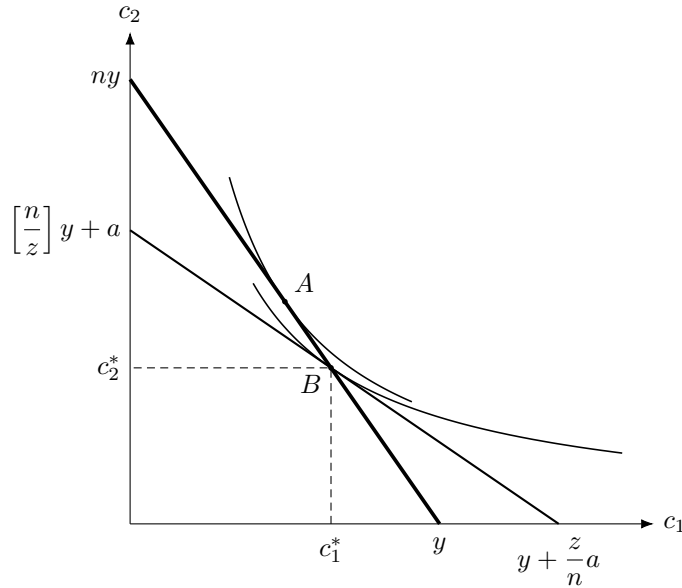


Figure 4.4: Growing population: even when the government sets $z = n$ so that prices are constant, the flatter budget line $c_1 + \frac{z}{n}c_2 \leq y$ cannot coincide with the steeper feasible-set line $c_1 + \frac{1}{n}c_2 \leq y$, leaving the equilibrium (c_1^*, c_2^*) at B below the efficient point A .

The striking lesson of Figure 4.4 is that even the non-inflationary policy $z = n$ fails to make the budget set coincide with the feasible set. The budget set tells a person that goods are equally available across periods — its slope reflects the constant price level. But that is not the true state of a *growing* economy. If each generation of young consumes one less good when young, there are n extra goods available for the old in each generation. With constant prices, the budget set never communicates this n -for-one technology of growth. Consequently people value future consumption *less* than they should, consuming too much when young and too little when old. Stable prices remove the inflation distortion but not the deeper distortion of population growth.

4.5 Government Expenditure

So far the printed money was returned to the old as a subsidy. We now let the government *keep* the goods it buys with seigniorage, so that we can study the cost of raising revenue in isolation from any benefit of spending it.

The government's budget constraint, in dollars, balances expenditure against revenue:

$$\begin{aligned}\text{Expenditure: } & p_t G_t + (1+i) D_{t-1}, \\ \text{Revenue: } & T_t + D_t + (M_t - M_{t-1}),\end{aligned}$$

where G_t is real government purchases, while D_t (debt), T_t (taxes) and M_t (money) are in dollars. The term $(1+i)D_{t-1}$ is principal plus interest on maturing debt, and $(M_t - M_{t-1})$ is the seigniorage implicitly levied by expanding the money supply. Multiplying through by $v_t = 1/p_t$ to express everything in goods,

$$G_t + (1+i) D_{t-1} v_t = v_t (T_t + D_t) + v_t (M_t - M_{t-1}).$$

For simplicity, suppose the government runs no debt and levies no explicit taxes, financing all of its purchases by seigniorage alone:

$$G_t = v_t (M_t - M_{t-1}) = v_t M_t \left(1 - \frac{1}{z}\right).$$

Denote constant real purchases per *old* person by

$$g = \frac{G_t}{N_{t-1}}.$$

We assume the goods the government acquires are used in a way that does not affect any individual's consumption choice and does not change the relative desirability of c_1 versus c_2 — for concreteness, imagine the government simply dumps them into the ocean. There is *no subsidy* to the old this time. This isolates the cost of raising revenue from the benefits of spending it.

The feasible set for stationary allocations now subtracts the government's take:

$$\begin{aligned}\text{Feasible set: } & N_t c_1 + N_{t-1} c_2 + G_t \leq N_t y, \\ \text{Per-capita form: } & c_1 + \frac{c_2}{n} + g \leq y, \\ \text{Constant population: } & c_1 + c_2 + g \leq y.\end{aligned}$$

Take constant population, so $v_{t+1}/v_t = 1/z$, and the lifetime budget constraint under stationarity is (with no subsidy, $a = 0$)

$$c_1 + z c_2 \leq y.$$

As before, the budget line is flatter — an individual must give up more c_1 to gain a unit of c_2 because money's return is low. But now the budget set has also *shrunk*: it lies strictly inside the no-inflation budget set, because the goods extracted by money creation are consumed by the government rather than returned to individuals. Both effects appear in Figure 4.5.

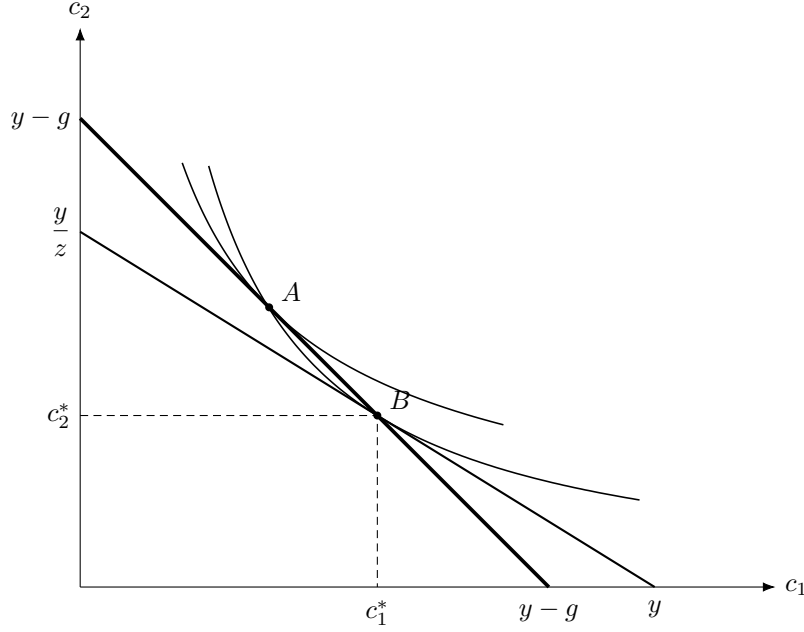


Figure 4.5: Government expenditure financed by seigniorage: the inflation-tax budget line $c_1 + zc_2 \leq y$ (intercept y/z) is both flatter and inside the feasible set $c_1 + c_2 \leq y - g$, so the inflation tax still distorts the consumption choice.

The picture makes plain that seigniorage finance distorts: the equilibrium bundle no longer lies on the feasible-set boundary, because the flatter budget line pushes people to hold too little money. The inflation tax buys the government g goods per old person, but at a deadweight cost.

4.6 A Non-Distorting Tax

Can the government raise the same revenue g without distorting the budget set? The distortion came entirely from the inflation tax falling on an economic activity — holding money — and thereby discouraging it. A tax that does not depend on any choice the individual makes would avoid this. That is a lump-sum tax.

Definition 4.4: Lump-Sum Tax

A *lump-sum tax* is a fixed levy τ of goods, collected from each old person, whose amount does not depend on any action the individual takes. Because it cannot be reduced by changing behavior, it leaves marginal incentives — in particular the incentive to hold money — undistorted.

Replace seigniorage with a lump-sum tax τ on the old. The two-period problem becomes

$$\begin{aligned} \text{Young: } c_{1,t} + v_t m_t &\leq y, \\ \text{Old: } c_{2,t+1} &\leq v_{t+1} m_t - \tau, \\ \text{Lifetime budget: } c_{1,t} + \frac{v_t}{v_{t+1}} c_{2,t+1} &\leq y - \frac{v_t}{v_{t+1}} \tau. \end{aligned}$$

Now hold both the population (hence money demand) and the money stock fixed over time, so $z = 1$ and $v_{t+1}/v_t = 1$: there is no inflation. Set the tax to fully finance the government, $\tau = g$. The lifetime budget constraint becomes

$$c_1 + c_2 \leq y - g,$$

which is *identical* to the per-capita feasible set $c_1 + c_2 + g \leq y$.

Lump-sum beats inflation

By using a lump-sum tax instead of seigniorage, the government raises the same real revenue g with *no distortion* of the budget set: the budget line coincides with the feasible-set line, so individuals are not induced to shrink their money balances to dodge the inflation tax. Any tax on a productive (positive) activity is inferior to a lump-sum tax, because the activity tax reduces the incentive to undertake the activity.

This is a recurring theme in public finance, and the monetary model delivers it in an especially clean form: the inflation tax is a tax on the useful service that money provides, and like any such tax it carries a deadweight loss that an equal-yield lump-sum tax avoids.

Remark (Where this is heading).

We have seen that money growth is an implicit, proportional, and distorting tax; that the revenue it raises is bounded by a Laffer-type relationship; and that an equal-yield lump-sum tax dominates it. Chapter 5 extends the monetary model to an open economy with more than one currency, and Chapter 6 turns to the institution that actually controls z in practice — the central bank.

Chapter 5

International Monetary Systems

So far each economy has had a single fiat money, issued by a single government, whose value was pinned down by the balance between the supply of money and the demand for real balances of its young (Chapters 1 and 4). The world, however, contains many governments, each printing its own currency. The organizing fact of international monetary economics is that these currencies trade for one another at a price: the *exchange rate*. A dollar buys a certain number of yen today and perhaps a different number tomorrow.

This chapter embeds the overlapping-generations model of money into a two-country world and asks what determines that price. We will find that the answer depends sharply on the institutions in place. When citizens may hold only their home currency, each country's money market clears separately and the exchange rate is determined by relative supplies and demands, just as a price should be. The moment we allow people to hold either currency, that determinacy collapses: a single world money market cannot pin down two currency values, and the exchange rate becomes whatever people believe it to be. The rest of the chapter studies why private portfolio diversification cannot anchor the rate, and the institutional devices — cooperative stabilization and unilateral defense — through which governments can, tracing who bears the cost of defending a peg.

5.1 International Exchange

To study these questions we extend the overlapping-generations setup of Chapter 1 to two countries, labeled a and b , each issuing its own fiat money. People live two-period lives in overlapping generations: they are endowed with goods when young, receive nothing when old, yet wish to consume in both periods. The two countries produce the *same* good — a unit of the good in country a is indistinguishable from a unit in country b — and people are indifferent to where the goods they buy were produced. We use superscripts a and b to tag the parameters and variables of each country. As before, all changes in a country's fiat money stock are used to buy goods for its government, and we assume free international trade in goods.

The two monies can be traded at the exchange rate e_t , defined as the number of units of country b money that one unit of country a money buys.

Definition 5.1: Exchange Rate

The *exchange rate* e_t is the units of country b money purchased with one unit of country a money. Taking country a to be the United States and country b to be Japan,

$$e_t = \frac{\text{Japanese yen}}{\text{U.S. dollar}},$$

the number of yen per dollar — equivalently, the number of yen a dollar can buy.

By definition, the owner of one unit of country a money at time t can buy v_t^a goods with it, and the owner of one unit of country b money can buy v_t^b goods. When people are free to trade currencies at the rate e_t , the holder of a unit of country a money has two options: buy v_t^a goods directly, or first trade the unit for e_t units of country b money and then buy $e_t v_t^b$ goods. The same comparison, reversed, faces a holder of country b money. Holders of both monies are indifferent between their two options — and arbitrage leaves no profit on the table — if and only if the two routes deliver the same goods:

$$v_t^a = e_t v_t^b \quad \Longleftrightarrow \quad e_t = \frac{v_t^a}{v_t^b} \left(= \frac{p_t^b}{p_t^a} \right).$$

This is the *Law of One Price*: the exchange rate equals the ratio of the goods-values of the two monies, or equivalently the ratio of the two countries' price levels.

Definition 5.2: Law of One Price

With free trade in goods and currencies, the exchange rate is the ratio of the real values of the two monies,

$$e_t = \frac{v_t^a}{v_t^b} = \frac{p_t^b}{p_t^a},$$

so that one unit of country a money buys the same quantity of goods whether spent at home or converted and spent abroad.

We now ask how this exchange rate behaves under alternative international monetary arrangements. Two polar cases organize the discussion: one in which young people are constrained to hold only the money of their home country, and one in which young people may hold any country's currency.

5.2 Foreign Currency Controls

Under *foreign currency controls*, the citizens of each country are permitted to carry over time only the fiat money of their own country. These controls do not forbid trade between the two countries: young people simply do not migrate, while an old citizen who wishes to buy foreign goods may travel abroad, exchange his money, and make the purchase. There is a single consumption good in the world. The decisive consequence of foreign currency controls is that each country now has its own, self-contained money supply and demand, which independently determine the value of its fiat money. The two money-market-clearing

conditions are

$$\begin{aligned} v_t^a M_t^a &= N_t^a (y^a - c_{1,t}^a), \\ v_t^b M_t^b &= N_t^b (y^b - c_{1,t}^b), \end{aligned}$$

each saying that the real value of a country's money stock equals the real balances its young generation wishes to hold (their endowment y net of first-period consumption c_1 , summed over the N_t young).

In the absence of market frictions, competition forces the price of the good to be equal across countries. Applying the Law of One Price, the exchange rate is therefore

$$e_t = \frac{v_t^a}{v_t^b} = \frac{N_t^a (y^a - c_{1,t}^a) / M_t^a}{N_t^b (y^b - c_{1,t}^b) / M_t^b} = \frac{N_t^a (y^a - c_{1,t}^a)}{N_t^b (y^b - c_{1,t}^b)} \cdot \frac{M_t^b}{M_t^a}.$$

What pins down the exchange rate under controls

The exchange rate — the value of country a money in terms of country b money — depends only on the *relative* levels of money demand and money supply in the two countries. A larger money demand in country a , or a smaller money supply, makes country a money more valuable and raises e_t .

Because each v_t evolves exactly as in the one-country model of Chapter 1, the exchange rate's growth rate follows from the two countries' growth rates of population, n , and of the money stock, z . Using $v_{t+1}/v_t = n/z$ in each country,

$$\frac{e_{t+1}}{e_t} = \frac{v_{t+1}^a/v_t^a}{v_{t+1}^b/v_t^b} = \frac{n^a/z^a}{n^b/z^b} = \frac{n^a}{n^b} \cdot \frac{z^b}{z^a}.$$

The exchange rate appreciates for country a when its population grows faster (raising its money demand) or when it prints money more slowly than country b .

5.3 Indeterminacy of the Exchange Rate

Foreign currency controls force people to swap monies before they can buy another country's goods, and in a world where exchanging money is itself costly, this is a real burden on international trade. Suppose instead that young people are free to hold and use the money of either country.

Once people may hold either money, we can no longer treat the two money markets separately; we must look at the *world* supply of and demand for money. The world supply of fiat money, measured in goods, is $v_t^a M_t^a + v_t^b M_t^b$, and the world demand for real balances is $N_t^a (y^a - c_{1,t}^a) + N_t^b (y^b - c_{1,t}^b)$. Market clearing requires

$$v_t^a M_t^a + v_t^b M_t^b = N_t^a (y^a - c_{1,t}^a) + N_t^b (y^b - c_{1,t}^b).$$

A serious problem now surfaces. We have a single equation but *two* unknowns, v_t^a and v_t^b . Such an equation has infinitely many solutions. Since $e_t = v_t^a/v_t^b$, world supply can be made to equal world demand at *any* positive exchange rate e_t : pick any $e_t > 0$, and there is a pair (v_t^a, v_t^b) with $v_t^a = e_t v_t^b$ that clears the world market.

Theorem 5.3: Exchange-Rate Indeterminacy

When citizens may hold either country's money, the single world money-market-clearing condition

$$v_t^a M_t^a + v_t^b M_t^b = N_t^a (y^a - c_{1,t}^a) + N_t^b (y^b - c_{1,t}^b)$$

does not determine the two currency values v_t^a, v_t^b separately. Every positive exchange rate $e_t = v_t^a / v_t^b$ is consistent with a world equilibrium. The exchange rate is *indeterminate*.

Proof. Fix any candidate rate $e_t > 0$ and impose $v_t^a = e_t v_t^b$. Substituting into the world-clearing condition gives

$$v_t^b (e_t M_t^a + M_t^b) = N_t^a (y^a - c_{1,t}^a) + N_t^b (y^b - c_{1,t}^b),$$

a single equation that solves for a unique positive v_t^b (and hence $v_t^a = e_t v_t^b$). Since this works for every $e_t > 0$, no economic condition selects among them. $\square$

This indeterminacy did not arise under foreign currency controls. There, the two separate market-clearing equations fixed the value of fiat money in each country, and their ratio then fixed the exchange rate. The reason for the change is economic, not merely algebraic. Because people are now free to hold either money, the size of one nation's money demand affects the real value of the *world* money supply but no longer determines the exchange rate, since residents are not restricted to their own currency. Symmetrically, the quantity of money printed by any one country does not determine the rate either, because that money can be spent anywhere.

Beliefs pin down what fundamentals do not

With no government action fixing the rate, the exchange rate in a unified world economy can be whatever people *believe* it to be. If those beliefs fluctuate, so will the exchange rate, because nothing else anchors it — and these fluctuations need not correspond to any change in real economic conditions.

5.4 International Currency Traders

Exchange-rate indeterminacy means a whole continuum of rates is consistent with a stationary equilibrium, so the rate may swing sharply as large holders rebalance their money portfolios. These swings make each individual currency a risky asset. Someone with access to only one currency watches the real value of his balances — and hence his old-age consumption — rise and fall with the exchange rate. A multinational, by contrast, can insure against this risk by holding a balanced portfolio of both monies: when the exchange rate moves, the loss on one currency is offset by the gain on the other.

To formalize this, consider an economy with three types of people:

1. citizens of country a , required by law to hold only country a money;
2. citizens of country b , required by law to hold only country b money;
3. multinational people, free to hold either currency.

Let N_t^a, N_t^b, N_t^c denote the numbers of people of each type in a generation born in period t .

Each country's money is held by its own citizens and, possibly, by multinationals. Let λ_t be the fraction of the multinationals' money balances held in the form of country a money. The two currency markets then clear according to

$$\begin{aligned} v_t^a M_t^a &= N_t^a (y^a - c_{1,t}^a) + \lambda_t N_t^c (y^c - c_{1,t}^c), \\ v_t^b M_t^b &= N_t^b (y^b - c_{1,t}^b) + (1 - \lambda_t) N_t^c (y^c - c_{1,t}^c). \end{aligned}$$

The more the multinationals wish to hold country a money — the larger λ_t — the greater the value of country a money and the smaller that of country b money, and hence the higher the exchange rate e_t . But precisely because multinationals may split their balances in any proportion they like, λ_t is free, and with it a whole range of equilibrium exchange rates. Indeterminacy persists; the multinationals' portfolio choice merely relabels it.

Remark (A frictionless idealization).

In practice, the diversification that frees a multinational from exchange-rate risk is not costless. Holding perfectly balanced stocks of two currencies, and continually rebalancing them, can be bothersome and expensive — one reason real-world agents do not hold the textbook-optimal hedged portfolio.

5.5 Fixed Exchange Rate

A *fixed exchange rate* is an equilibrium in which the rate is constant over time. From the growth law derived under currency controls, constancy requires

$$\begin{aligned} e_{t+1} &= e_t \\ \implies z^a &= \frac{n^a}{n^b} z^b. \end{aligned}$$

To hold the rate fixed, then, one or both countries must choose money-creation rates satisfying this relation. The cost is a loss of monetary independence: a government committed to a fixed exchange rate can no longer set its money-growth rate freely to extract a chosen amount of seigniorage.

The peg–seigniorage trade-off

A country may set its rate of money creation either to fix the exchange rate ($z^a = (n^a/n^b) z^b$) or to raise its preferred level of seigniorage revenue, but *not both*. Committing to a peg surrenders the seigniorage instrument.

5.5.1 Cooperative Stabilization

Cooperative stabilization refers to exchange-rate policy run jointly by two governments that stand ready to exchange their currencies at a given rate. The rate is then fixed over time. Note that in the absence of foreign currency controls, currencies are held only voluntarily — and no currency will be held voluntarily if its value is expected to fall over time relative to the others. A depreciating currency offers a lower rate of return, which would induce everyone to abandon it for the alternatives. A credible peg, backed by both governments, removes that asymmetry and lets both monies be held side by side.

5.5.2 Unilateral Defense

Getting every country to agree on every exchange rate is difficult; sovereign nations rarely commit without limit. A fixed rate can nonetheless be supported without the full cooperation of the foreign central bank, but only by a costly device: the government doing the defending must tax its own citizens and use the proceeds to buy up the foreign currency that the market demands.

To see how a *unilateral defense* works, return to the two-country economy with no foreign currency controls and no cooperation between central banks. Suppose the entire world arbitrarily decides to convert part of its country *a* money into country *b* money. Under cooperative stabilization the holders of country *a* money would simply exchange it for country *b* money supplied by country *b*'s central bank. Under unilateral defense, by contrast, country *b* refuses to trade one money for another — though its central bank will still trade money for goods. The government of country *a* then pledges to tax its old in order to defend the fixed rate. (The tax falls on the old because they are the citizens who lose if the nation's money loses value.) Because there are no currency controls, some of each money is held by the old of each country. The world currency market is

$$\begin{aligned} v_t^a M_t^a + v_t^b M_t^b &= N_t^a (y^a - c_{1,t}^a) + N_t^b (y^b - c_{1,t}^b) \\ \iff \bar{e} v_t^b M_t^a + v_t^b M_t^b &= N_t^a (y^a - c_{1,t}^a) + N_t^b (y^b - c_{1,t}^b), \end{aligned}$$

where $\bar{e}$ denotes the fixed exchange rate and the second line uses $v_t^a = \bar{e} v_t^b$.

The contrast between the two policies is sharpest in a concrete numerical example, which we now work through.

Example (Defending a peg: cooperation versus unilateral defense).

Let countries *a* and *b* be identical. In each, every generation has population 100, and each young person wants real balances worth 10 goods, so aggregate real balances in each country are

$$N_t^a (y^a - c_{1,t}^a) = N_t^b (y^b - c_{1,t}^b) = (100)(10) = 1000.$$

The total fiat money stock is \$800 in country *a* and £600 in country *b*. There are no currency controls, and each money is held in both countries: the stocks are spread evenly across the 200 initial old (100 born per generation in each country), so each member of the initial old holds \$4 and £3, regardless of citizenship. The rate is fixed at $\bar{e} = \frac{1}{2}$; that is, \$1 trades for £0.5.

Baseline. From the world clearing condition $\bar{e} v_t^b M_t^a + v_t^b M_t^b = 2000$,

$$\frac{1}{2} v_t^b(800) + v_t^b(600) = 1000 + 1000 \implies v_t^b = 2, \quad v_t^a = \bar{e} v_t^b = 1.$$

Each old person's consumption is the real value of his total money holdings,

$$c_t^a = c_t^b = v_t^a \cdot 4 + v_t^b \cdot 3 = 1 \cdot 4 + 2 \cdot 3 = 10 \text{ goods.}$$

Cooperative stabilization. Now suppose every member of the initial old halves his real balances of country *a* money, turning in \$2 each to acquire country *b* money. Country *b*'s authority cooperates, printing as much currency as demanded and shipping it to country *a*. At the fixed rate $\frac{1}{2}$, country *b* prints £0.5 for every dollar turned in. With 200 old each turning in \$2, the dollar stock falls by \$400 and the pound stock rises by £200, leaving $M_t^a = 400$ and $M_t^b = 800$. Re-solving,

$$\frac{1}{2} v_t^b(400) + v_t^b(800) = 2000 \implies v_t^b = 2, \quad v_t^a = \bar{e} v_t^b = 1,$$

so $c_t^a = c_t^b = 1 \cdot 4 + 2 \cdot 3 = 10$ goods. Consumption is unchanged: world money demand is unchanged, and the real world money supply is unchanged because the old simply hold fewer dollars and more pounds.

Unilateral defense. Suppose instead country *b* refuses to print pounds to accommodate the old, and country *a* honors its pledge by an equal tax on each of its old citizens, using the proceeds to supply the country *b* money demanded. The steps are as follows. There are 200 old across the two countries, each turning in \$2, so country *a*'s authority must acquire £200 in exchange (at $\bar{e} = \frac{1}{2}$). Since country *b* prints nothing, country *a* must *give up goods* equal in value to the pounds it acquires, namely $v_t^b(\text{£}200)$ goods. Because country *a* can tax only its own 100 old citizens, each pays a tax of

$$\frac{200 v_t^b}{100} = 2v_t^b.$$

To evaluate the tax we need v_t^b . Defending the rate this way shrinks country *a*'s money stock to 400 while country *b*'s stays at $M_t^b = 600$:

$$\frac{1}{2} v_t^b(400) + v_t^b(600) = 2000 \implies v_t^b = 2.5, \quad v_t^a = \bar{e} v_t^b = 1.25.$$

By decreasing the total world money supply while leaving money demand unchanged, the unilateral defense *raises* the value of every currency — in sharp contrast to cooperative stabilization, where each currency's value was unchanged. The tax per old citizen of country *a* is therefore

$$2v_t^b = 2(2.5) = 5 \text{ goods.}$$

The collected goods, 500 in total, buy newly issued pounds: at 2.5 goods per pound the 500 goods (shared by the 200 old returning country *a* money) purchase £200, so each old person acquires one additional pound. After turning in \$400 (leaving \$2 each) and receiving the new pounds, each old person who began with £3 ends with £4.

For the old of country b , who pay no tax, old-age consumption is

$$c_t^b = v_t^a \cdot 2 + v_t^b \cdot 4 = 1.25 \cdot 2 + 2.5 \cdot 4 = 12.5 \text{ goods.}$$

For the old of country a , who must pay the tax to defend their currency,

$$c_t^a = v_t^a \cdot 2 + v_t^b \cdot 4 - (\text{tax}) = 12.5 - 5 = 7.5 \text{ goods.}$$

The example delivers a clean distributional verdict. The old of country b *benefit* from country a 's unilateral defense: the real value of their currency holdings rises and they pay no tax. The old of country a are made *worse off* than under cooperative stabilization. In effect, only the citizens of the defending country pay the tax, and that tax raises the value of money held by everyone — so the policy transfers wealth from the taxpayers of the country defending the exchange rate to the money holders of the other country.

Who pays for a unilateral peg

A unilateral defense shrinks the world money supply by taxing the defending country's old, which raises every currency's real value. The defending country's old bear the entire cost; the foreign country's old enjoy a windfall. Cooperative stabilization, by reallocating currencies without changing the world money supply, imposes no such cost.

Chapter 6

The Central Bank and Its Independence

The earlier chapters treated the money supply as something the monetary authority simply chooses: a single growth factor z in the inflation model of Chapter 4, a lever pulled at will to collect seigniorage. But who is this authority, and why should we trust it to pull the lever wisely rather than opportunistically? Behind the clean letter z stands an actual institution—in the United States, the Federal Reserve System—with a deliberate internal architecture and a contested degree of insulation from the elected government.

This chapter steps away from formal modelling to describe that institution. We first map the structure of the Federal Reserve System: the twelve regional Reserve Banks, the Board of Governors, and the Federal Open Market Committee through which monetary policy is actually conducted. We then turn to the central political question of monetary economics—*central bank independence*—asking what independence means, why a society might want its central bank insulated from day-to-day politics, and what it gives up in exchange. The European Central Bank serves as a contrasting design at the end. The recurring theme is that the institutions are engineered to balance two goals in tension: technical competence shielded from short-run political pressure, and democratic accountability.

6.1 The Federal Reserve System

The Federal Reserve System is not a single bank but a federated structure with several distinct organs, each with its own membership and powers. The system comprises:

- the twelve regional *Federal Reserve Banks*;
- the *Board of Governors* of the Federal Reserve System, in Washington, D.C.;
- the *Federal Open Market Committee* (FOMC), which conducts open market operations;
- the *Federal Advisory Council*; and
- roughly 2,900 member commercial banks.

The design distributes power deliberately. Regional banks anchor the system in local economic conditions and are partly owned by private commercial banks; the Board of Governors is a public body appointed by the federal government; and the FOMC blends the two. We take each organ in turn.

6.1.1 Federal Reserve Banks

The twelve Federal Reserve Banks are *quasi-public* institutions: each is owned by the private commercial banks in its district that are members of the Federal Reserve System, yet each carries out public functions and is overseen by the Board of Governors. This hybrid ownership—private in form, public in mandate—is the first place the system’s balancing of interests appears.

Each Reserve Bank has nine directors. Member banks elect six of them, and the Board of Governors appoints the remaining three. The nine seats are organized into three classes, each designed to represent a different constituency:

- **Class A** (three directors, elected by member banks): professional bankers;
- **Class B** (three directors, elected by member banks): prominent leaders from industry, labor, agriculture, or the consumer sector;
- **Class C** (three directors, appointed by the Board of Governors): persons who may *not* be officers, employees, or stockholders of banks, so as to represent the broader public.

The nine directors together appoint the president of their Reserve Bank, subject to approval by the Board of Governors. The Class structure is a microcosm of the whole system’s intent: bankers are at the table, but they do not sit alone, and a public-interest bloc is built in by construction.

Functions of the Reserve Banks

The Reserve Banks perform the operational work of central banking within their districts. Their functions include:

- clearing checks;
- issuing new currency and withdrawing damaged currency from circulation;
- administering and making discount loans to banks in their districts;
- evaluating proposed bank mergers and applications for banks to expand their activities;
- acting as liaisons between the business community and the Federal Reserve System;
- examining bank holding companies and state-chartered member banks;
- collecting data on local business conditions; and
- employing staffs of professional economists to research questions bearing on the conduct of monetary policy.

The Special Role of the Federal Reserve Bank of New York

Among the twelve, the Federal Reserve Bank of New York occupies a distinctive position, for four reasons.

1. Its district contains many of the largest commercial banks in the United States, whose safety and soundness are paramount to the health of the national financial system.
2. It is actively involved in the bond and foreign exchange markets, where open market operations and intervention are physically carried out.
3. It is the only Federal Reserve Bank that is a member of the Bank for International Settlements (BIS).
4. Its president is the only Reserve Bank president who is a *permanent* voting member of the FOMC, serving as the committee's vice-chair.

Together with the chair and vice-chair of the Board of Governors, the president of the New York Fed is therefore one of the three most important officials in the entire system.

The Reserve Banks and Monetary Policy

Although the headline policy decisions are made centrally, the Reserve Banks retain real instruments. Their directors “establish” the discount rate (subject to review by the Board of Governors), decide which banks may obtain discount loans, and select one commercial banker from each district to serve on the Federal Advisory Council, which consults with the Board of Governors and supplies information used in conducting policy. Most importantly, five of the twelve Reserve Bank presidents hold a vote on the FOMC at any given time.

Member Banks

Membership in the Federal Reserve System is partly mandatory and partly voluntary. All *national* banks are required to be members; commercial banks chartered by states are not required to join but may choose to. The Depository Institutions Deregulation and Monetary Control Act of 1980 narrowed the practical difference between the two: it subjected *all* depository institutions to the same reserve requirements as member banks and gave all of them access to Federal Reserve facilities.

6.1.2 The Board of Governors

The Board of Governors is the public, central organ of the system, headquartered in Washington, D.C. Its design is engineered to insulate it from short-run political pressure—the structural foundation of the independence debate that occupies the second half of this chapter.

Structure of the Board of Governors

- **Seven members**, headquartered in Washington, D.C.
- Each is **appointed by the President** and **confirmed by the Senate**.
- Each serves a single **14-year, non-renewable term**.

- Members are required to come from different Federal Reserve districts.
- The **Chairman** is chosen from among the governors and serves a four-year term.

The long, non-renewable term is the key feature. Because a governor cannot be reappointed, he or she has no incentive to court the appointing President for a second term; and because the terms are staggered and outlast any single presidency, no one administration can pack the Board quickly. We return to this design in Section 6.2.

Duties of the Board of Governors

The Board's powers reach across the whole system and the banking sector. It:

- votes on the conduct of open market operations (as part of the FOMC);
- sets reserve requirements;
- controls the discount rate through a “review and determination” process;
- sets margin requirements (the minimum down payment on securities bought on credit);
- sets the salaries of the president and officers of each Reserve Bank and reviews each bank's budget;
- approves bank mergers and applications for new activities;
- specifies the permissible activities of bank holding companies; and
- supervises the activities of foreign banks operating in the United States.

The Chairman

The Chairman of the Board of Governors is the public face of the system. The Chairman advises the President on economic policy, testifies before Congress, speaks for the Federal Reserve System to the media, and may represent the United States in negotiations with foreign governments on economic matters.

6.1.3 The Research Staff

The Federal Reserve System is the largest employer of economists not merely in the United States but in the world. The central task of these economists is to follow the incoming economic data from government agencies and private-sector organizations, and to advise policymakers on the likely direction of the economy and the probable impact of monetary policy actions. The seemingly mechanical lever z of Chapter 4 is thus pulled only after a large professional apparatus has weighed in on where it should be set.

6.1.4 The Federal Open Market Committee

Open market operations are the principal tool of U.S. monetary policy, and they are governed by the Federal Open Market Committee.

The FOMC

- Meets **eight times a year**.
- Consists of the **seven** members of the Board of Governors, the president of the Federal Reserve Bank of New York, and the presidents of **four** of the other eleven Reserve Banks (who vote on a rotating basis).
- The Chairman of the Board of Governors also chairs the FOMC.
- Issues directives to the trading desk at the Federal Reserve Bank of New York, where the operations are executed.

The arithmetic of the votes is itself a statement of where power lies: the seven governors plus the New York Fed president are permanent voters, so the Washington-appointed bloc commands a structural majority, while the rotating four regional presidents keep the rest of the country represented.

6.2 The Independence of Central Banks

We now turn to the question that gives this chapter its title. A central bank is *independent* to the extent that it can set monetary policy without being overridden by the elected government. Independence is not all-or-nothing, and a first step is to distinguish two kinds.

Definition 6.1: Instrument vs. Goal Independence

A central bank has *goal independence* if it can set the *objectives* of monetary policy (for example, choosing the target rate of inflation). It has *instrument independence* if, given objectives set elsewhere, it can choose the *tools* used to pursue them (open market operations, the discount rate, reserve requirements) free of political interference. A central bank may have one without the other.

6.2.1 Gauging the Fed's Independence

Several structural features bear on how independent the Federal Reserve actually is, and they cut in both directions.

Sources of independence.

- **Independent revenue.** The Fed funds itself from the interest on its securities holdings and from fees, so it does not depend on annual appropriations from Congress. A body that need not pass the hat is far harder to discipline.
- **Long, staggered, non-renewable terms** on the Board of Governors, as described above, blunt the appointing President's leverage.

Sources of dependence.

- **The Fed's structure is written by Congress and is subject to change at any time.** The Federal Reserve was created by statute and can be reorganized—or its powers curtailed—by statute. Its independence exists at the sufferance of the legislature.
- **Presidential influence.** The President shapes the Fed indirectly through influence on Congress, and directly by appointing the members of the Board of Governors and designating its Chairman. Because the governors' fourteen-year terms are not concurrent with the presidency, this influence accrues only slowly—but over time it is real.

Remark (Independence is granted, not inherent).

The crucial point is that the Fed's independence is a *policy choice embedded in legislation*, not a constitutional given. Congress could revoke it. The interesting economics, then, is not whether the Fed is independent in some absolute sense, but why a legislature would *choose* to tie its own hands by creating an authority it does not directly control. The next subsection takes up that question.

6.2.2 The Case For and Against Independence

Arguments for independence

- **Inflationary bias.** Subjecting monetary policy to direct political pressure would impart an inflationary bias: elected officials facing reelection have a standing temptation to stimulate the economy now and let the inflation arrive later.
- **The political business cycle.** Without insulation, policymakers may engineer booms to coincide with elections, generating an artificial cycle in output and inflation timed to the political calendar rather than to economic fundamentals.
- **Deficit accommodation.** A politically controlled central bank could be pressed into financing large budget deficits by creating money to buy government debt—the *accommodation* that, in the inflation model of Chapter 4, raises the money-growth factor z and with it the inflation tax. Independence is a commitment device against this temptation.
- **The principal–agent problem.** Monetary policy is too important, and too technical, to be run on short electoral horizons. The principal–agent problem—the gap between what is good for the public and what is good for the official—is arguably worse for elected politicians, whose horizon ends at the next election, than for insulated technocrats.

Arguments against independence

- **Undemocratic.** A powerful body that is not elected exercises enormous influence over the economy, which sits uneasily with democratic principles.
- **Unaccountable.** If the central bank performs poorly, there is no straightforward electoral mechanism to remove those responsible.
- **Coordination.** Independence makes it harder to coordinate fiscal and monetary policy, which may pull in opposite directions when they ought to act together.

- **Mixed record.** The Fed has not always used its independence successfully—insulation from politics is no guarantee of good policy.

The independence trade-off

Independence buys a *credible commitment* to low and stable inflation by removing monetary policy from the short electoral horizon—it is, in effect, a device for tying the government’s hands against the inflationary bias of Chapter 4. The price is a loss of direct democratic accountability and of easy fiscal-monetary coordination. The institutional design surveyed in Section 6.2 is the attempt to purchase the first without paying too much of the second.

6.2.3 The European Central Bank

The European Central Bank (ECB) offers an instructive contrast in institutional design. Like the Federal Reserve it is a federated system, but the balance between the center and the national components is struck differently.

Distinctive features.

- The *National Central Banks* control their own budgets and, collectively, the budget of the ECB itself—decentralization runs deeper than in the U.S. system.
- Monetary operations are not centralized; they are carried out by the national central banks rather than from a single desk.
- The ECB does not supervise and regulate financial institutions, a function that in the United States the Fed performs prominently.

The Governing Council. The ECB’s policy is set by its Governing Council, the analogue of the FOMC.

- It holds monthly meetings at the ECB in Frankfurt, Germany.
- It comprises the heads of the National Central Banks together with the six members of the Executive Board.
- It operates by consensus; the ECB announces the target rate and takes questions from the media.
- To keep the council at a workable size as new countries join the euro area, voting proceeds on a system of rotation.

Independence. By statutory design the ECB is among the most independent central banks in the world.

- Members of the Executive Board serve long terms.
- It determines its own budget.

- Its overriding mandate is *price stability*, which makes it *less goal-independent* than the Fed: the goal is fixed for it by treaty, so its independence is largely about instruments rather than objectives.
- Most strikingly, its charter cannot be changed by ordinary legislation; it can be amended only by revising the Maastricht Treaty—a far higher bar than the statute that governs the Fed.

Remark (Two designs, one trade-off).

The Fed and the ECB sit at different points on the goal–instrument spectrum. The ECB has a single, treaty-fixed objective (price stability) but exceptionally strong protection of its instruments, while the Fed enjoys greater goal latitude—its mandate also embraces maximum employment—but rests on a charter that Congress can rewrite. Both are answers to the same underlying problem: how to make a credible commitment to sound money while remaining, in the end, answerable to the public. With the architecture of the monetary authority in place, Chapter 7 turns to the mechanics by which that authority actually changes the quantity of money.

Chapter 7

The Money Supply Process

The previous chapter described the central bank as an institution—who sits on its board, how independent it is, who pulls the lever. This chapter asks the mechanical question that the institutional account left open: when the central bank decides to change the quantity of money, how does it actually do so? The abstract growth factor z of the inflation model in Chapter 4 has to be implemented through real transactions on a real balance sheet, and the quantity of money the public ends up holding is not chosen by the central bank alone.

In fact three sets of actors jointly determine the money supply: the central bank (in the United States, the Federal Reserve System, or “the Fed”), the commercial banks that take deposits and make loans, and the depositors who decide how much of their money to hold as cash rather than as bank deposits. We proceed in two stages. First we follow the Fed’s own balance sheet and show how open market operations and discount loans change the *monetary base*—the raw material of the money supply. Then we show how the banking system multiplies that base into a much larger stock of money, and how the behavior of banks and depositors modifies the size of the multiplier. The recurring theme is that the Fed controls the base tightly but the multiplier only loosely, so the final money supply is the product of decisions made by all three players.

7.1 The Players and the Factors

Three players in the money supply process

1. **The central bank**—the Federal Reserve System (the Fed)—which conducts monetary policy.
2. **Banks**—depository institutions, i.e. financial intermediaries that take deposits and make loans.
3. **Depositors**—individuals and institutions that hold deposits at, and currency issued through, the banking system.

The Fed can change the money supply through two principal channels, plus a handful of secondary ones.

The two principal factors are *open market operations*—the Fed’s purchase or sale of securities in the open market—and *loans to financial institutions*, the discount loans the Fed extends to banks. Both, as we will see, inject reserves into or drain reserves out of the banking system.

Several *other factors* also move the money supply, though they are not deliberate policy instruments:

- *float*—reserves created temporarily in the check-clearing process;
- *Treasury deposits at the Fed*—when the Treasury moves balances into its account at the Fed, reserves leave the banking system;
- *interventions in the foreign exchange market*—which, under a broad definition, are simply open market operations conducted in foreign rather than domestic assets.

7.2 The Fed’s Balance Sheet

Everything the Fed does to the money supply shows up on its balance sheet, so it pays to fix that balance sheet first. In its simplest form it has two assets and two liabilities.

Table 7.1: The Fed’s balance sheet in simplified form.

Assets	Liabilities
Securities	Currency in circulation
Loans to financial institutions	Reserves

Definition 7.1: The Monetary Base

The *monetary base* (also called high-powered money) is the sum of the Fed’s monetary liabilities:

$$MB = C + R,$$

where C is currency in circulation—cash in the hands of the public—and R is reserves—bank deposits held at the Fed plus vault cash. The monetary base is the raw material out of which the banking system creates the broader money supply.

The two liability items are worth distinguishing carefully, because only changes in them move the base.

- **Currency in circulation** is currency in the hands of the public. (Currency still sitting in the Fed’s or banks’ own vaults is not “in circulation.”)
- **Reserves** are banks’ deposits at the Fed together with their vault cash. Reserves are an asset of the banks but a liability of the Fed.

On the asset side:

- **Government securities** are the Fed’s holdings of (chiefly U.S. Treasury) bonds. Buying and selling them is how the Fed changes the money supply, and they earn the Fed interest along the way.

- **Discount loans** are the loans the Fed makes to banks. They supply reserves to the banking system and earn the Fed the discount rate.

Remark (The Fed and the PBoC compared).

The same two-sided logic describes any central bank, but the dominant entries differ across countries. For the People's Bank of China (PBoC), the major asset items are *foreign exchange* and *claims on other depository corporations*, and the major liability item is *reserve money*. For the Fed, the major assets are *U.S. Treasury securities* and *agency mortgage-backed securities (MBS)*, and the major monetary liability is *deposits held by depository institutions* (other than term deposits). That foreign exchange looms so large on the PBoC's books, but barely figures on the Fed's, reflects the very intervention mechanics discussed in Chapter 5.

7.3 Open Market Purchases

We now trace, step by step, what an open market purchase does to reserves and to the monetary base. To keep the bookkeeping transparent we follow a single purchase of \$100 million of securities, and—until we reach the multiplier—we assume a 100% required reserve ratio. The crucial lesson is that the effect on *reserves* depends on who sells the bonds and what they do with the proceeds, whereas the effect on the *monetary base* does not.

7.3.1 Purchase from a Bank

Suppose the Fed buys \$100 million of securities directly from a bank. The bank gives up securities and receives reserves; the Fed acquires securities and creates the reserves to pay for them. The two balance-sheet changes are shown below (T-accounts record only the *changes*, so each side need not list every item).

Table 7.2: Open market purchase from a bank: changes in T-accounts.

Federal Reserve System			
Assets		Liabilities	
Securities	+\$100m	Reserves	+\$100m

Banking System	
Assets	Liabilities
Securities	−\$100m
Reserves	+\$100m

The net result is that reserves rise by \$100 million, and therefore the monetary base rises by \$100 million, with *no* change in currency. The Fed has simply swapped an asset (securities) for a newly created liability (reserves).

7.3.2 Purchase from the Nonbank Public

Now suppose the Fed buys the securities not from a bank but from a member of the nonbank public—a household or a firm. The outcome now turns on what the seller does with the Fed's check.

Case 1: the seller deposits the check in a bank

If the seller deposits the Fed's check into a checking account, the funds re-enter the banking system. Tracing the three balance sheets:

Table 7.3: Open market purchase from the public, proceeds deposited: changes in T-accounts.

Nonbank Public			
Assets		Liabilities	
Securities	−\$100m		
Checkable deposits	+\$100m		

Banking System			
Assets		Liabilities	
Reserves	+\$100m	Checkable deposits	+\$100m

Federal Reserve System			
Assets		Liabilities	
Securities	+\$100m	Reserves	+\$100m

The effect on the Fed's balance sheet, and hence on reserves and the monetary base, is *identical* to the purchase from a bank: reserves and the base each rise by \$100 million.

Case 2: the seller cashes the check

If instead the seller cashes the Fed's check and walks away with currency, the proceeds never enter the banking system.

Table 7.4: Open market purchase from the public, proceeds taken as cash: changes in T-accounts.

Nonbank Public			
Assets		Liabilities	
Securities	−\$100m		
Currency	+\$100m		

Federal Reserve System			
Assets		Liabilities	
Securities	+\$100m	Currency in circulation	+\$100m

Now reserves are unchanged. Instead currency in circulation rises by the amount of the purchase. The monetary base still increases by the full \$100 million—but as currency rather than as reserves.

7.3.3 Summary of Open Market Purchases

Open market purchases and the monetary base

- The effect of an open market purchase on **reserves** depends on whether the seller of the bonds keeps the proceeds in deposits (reserves rise) or in currency (reserves unchanged).
- The effect of an open market purchase on the **monetary base** is the same regardless: the base always rises by the full amount of the purchase.

The asymmetry matters because, as the next section shows, only *reserves* feed the multiple expansion of deposits. Currency held by the public adds to the base dollar for dollar but is not multiplied.

7.4 Loans to Financial Institutions

The second principal factor is the discount loan. When the Fed lends \$100 million to a bank, it credits the bank's reserve account and records a loan as its own asset; the bank gains reserves and records a corresponding borrowing.

Table 7.5: A discount loan from the Fed: changes in T-accounts.

Banking System			
Assets		Liabilities	
Reserves	+\$100m	Loans (borrowing from Fed)	+\$100m

Federal Reserve System			
Assets		Liabilities	
Loans (borrowing from Fed)	+\$100m	Reserves	+\$100m

The monetary liabilities of the Fed have risen by \$100 million, so the monetary base rises by the same amount. The mechanics are parallel to an open market purchase: a new Fed asset (the loan, rather than securities) is matched by newly created reserves.

7.5 Multiple Deposit Creation

We have seen how the Fed changes the monetary base. We now turn to how the banking system turns a given base into a much larger money supply. Before building the model it is worth being clear about the limits of the Fed's control.

How tightly does the Fed control the base?

Open market operations are firmly controlled by the Fed. Discount loans are *not*: the Fed sets the discount rate, but it cannot dictate how much banks choose to borrow. It is therefore useful to split the base into a part the Fed controls and a part it does not,

$$MB_n = MB - BR,$$

where BR is borrowed reserves (discount loans outstanding) and MB_n is the *non-borrowed monetary base*. The money supply is positively related both to the nonborrowed base MB_n and to borrowed reserves BR .

Remark (Structural monetary policy).

The distinction is sharper in some systems than others. China's frequent use of *structural* monetary policy means that a large share of base money is supplied through targeted lending facilities whose volume is steered by government policy guidance rather than left to banks' discretion—closer in spirit to a controllable BR than the U.S. arrangement assumes.

7.5.1 The Simple Model

Start from the most idealized case: banks hold no excess reserves, and the public holds no currency. Then every dollar of reserves is held only because it is required against deposits, so total reserves R equal required reserves RR :

$$RR = R.$$

Required reserves are, by definition, the required reserve ratio r times the stock of checkable deposits D . Setting required reserves equal to total reserves and solving for deposits,

$$\begin{aligned} r \cdot D &= R \\ \implies D &= \frac{1}{r} R \\ \implies \Delta D &= \frac{1}{r} \Delta R. \end{aligned}$$

A \$1 increase in reserves supports a $1/r$ increase in deposits. With a 10% reserve ratio, for instance, each extra dollar of reserves can support ten dollars of deposits: the banking system multiplies reserves into money.

Remark (Critique of the simple model).

The simple model overstates the multiplier because it ignores every leakage that stops the expansion short:

- **Currency drains.** Whenever the public chooses to hold proceeds as cash rather than as deposits, those dollars leave the banking system; currency undergoes no multiple expansion.

- **Excess reserves.** Banks need not lend out every spare dollar; reserves a bank chooses to hold above the requirement are not relent and do not create new deposits.
- **Behavior matters.** Depositors' decisions about how much currency to hold, and banks' decisions about how many excess reserves to keep, therefore move the money supply independently of the Fed.

The advanced model puts these leakages in explicitly.

7.5.2 The Advanced Model

The full picture has all three players acting. Before the algebra, it helps to collect the comparative statics: which variable each player controls, and which way the money supply moves when that variable rises.

Table 7.6: Factors determining the money supply.

Player	Variable	Change	M response	Reason
Federal Reserve	Nonborrowed base MB_n	↑	↑	More base for deposit creation
	Required reserve ratio r	↑	↓	Less multiple deposit expansion
Banks	Borrowed reserves BR	↑	↑	More base for deposit creation
	Excess reserves	↑	↓	Fewer loans and less deposit creation
Depositors	Currency holdings	↑	↓	Less multiple deposit expansion

We now derive the multiplier that delivers these signs. Define money as currency plus checkable deposits—the monetary aggregate M_1 —and link the money supply M to the monetary base MB through the *money multiplier* m :

$$M = m \times MB.$$

In the idealized environment of the simple model, $m = 1/r$. To go beyond it, we let the public's desired currency holdings and the banks' desired excess reserves both grow in proportion to deposits.

Definition 7.2: Currency and excess-reserve ratios

Let C be currency held by the public and ER excess reserves held by banks. Assuming both grow proportionally with checkable deposits D , define the *currency ratio* and the *excess-reserve ratio* as

$$c \equiv \frac{C}{D}, \quad e \equiv \frac{ER}{D}.$$

The ratio c summarizes depositor behavior; e summarizes bank behavior; the required reserve ratio r is the Fed's instrument.

Total reserves are required reserves plus excess reserves, and required reserves are r times deposits, so

$$\begin{aligned} R &= RR + ER \\ &= r \cdot D + e \cdot D \\ &= (r + e) \cdot D. \end{aligned}$$

The monetary base is currency plus reserves:

$$\begin{aligned} MB &= C + R \\ &= c \cdot D + (r + e) \cdot D \\ &= (c + r + e) \cdot D. \end{aligned}$$

This expression says exactly how much base is needed to support the existing stock of deposits, currency, and excess reserves. Inverting it gives deposits as a function of the base, $D = MB/(c + r + e)$.

The total money supply is currency plus deposits, $M = C + D = (1 + c)D$. Substituting for D ,

$$\begin{aligned} M &= C + D \\ &= (1 + c) \cdot D \\ &= (1 + c) \cdot \frac{MB}{c + r + e} \\ &= \frac{1 + c}{c + r + e} \cdot MB. \end{aligned}$$

Comparing with $M = m \cdot MB$ identifies the multiplier.

Theorem 7.3: The money multiplier

With a currency ratio c , an excess-reserve ratio e , and a required reserve ratio r , the money multiplier is

$$m = \frac{1 + c}{c + r + e}.$$

The money supply is $M = m \cdot MB$.

The formula reproduces every sign in Table 7.6. A higher required reserve ratio r or a higher excess-reserve ratio e enlarges the denominator and shrinks the multiplier; a higher currency ratio c also shrinks it, because c enters the denominator more heavily than the numerator (currency is base that does not get multiplied). Raising the base MB —whether through the nonborrowed base or through borrowed reserves—raises M proportionally.

Remark (Recovering the simple model).

Set $c = e = 0$: the public holds no currency and banks hold no excess reserves. Then

$$m = \frac{1 + 0}{0 + r + 0} = \frac{1}{r},$$

exactly the simple-model multiplier. The advanced model is thus the simple model with two leakages—currency and excess reserves—restored. It makes precise the earlier point that the Fed controls the base far more tightly than it controls the multiplier: MB_n is

the Fed's lever, but c belongs to depositors and e to banks, so the final money supply is the joint product of all three players' decisions.